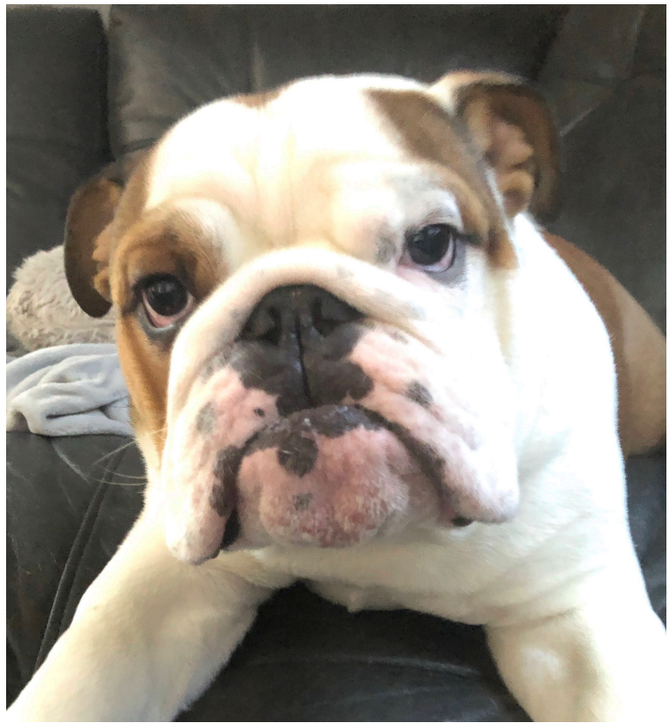
July 6, 2023



Written HW 1 : Due tonight by midnight

Reflection 2 : Due Friday by midnight

Conceptual HW 2 : Due Sunday by midnight

Lab : Tuesday

Written HW 2 : Due Tuesday by midnight

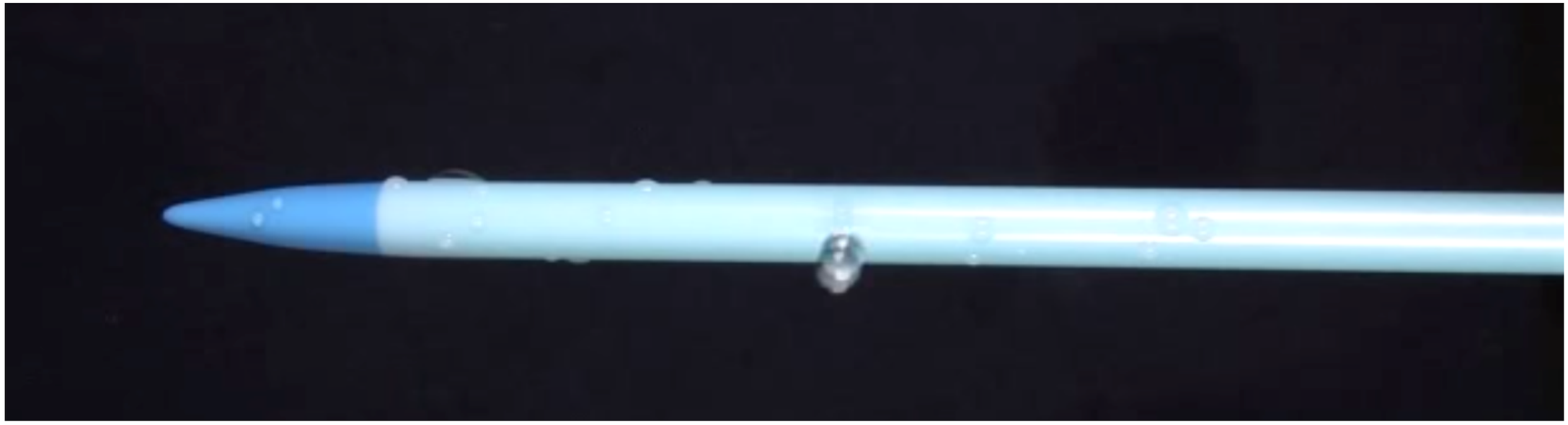
Office Hours

Mondays Noon-2pm
Thursdays 2-3pm

	Monday	Tuesday	Wednesday	Thursday	Friday
Week 2	July 3 No Class	July 4 No Classes!	July 5 V2 CH 5 HW	July 6 V2 CH 6 Studio : Field Calcs	July 7 V2 CH 6 Reflection
Week 3	July 10 V2 CH 6	July 11 Homework 2 Lab W3 - Using E	July 12 V2 CH 7	July 13 V2 CH 7 Studio : Complex Es	July 14 V2 CH 7 Reflection
Week 4	July 17 V2 CH 8	July 18 Homework 3 Lab W4 - Electric Potential	July 19 V2 CH 8	July 20 V2 CH 9 Studio : V and C	July 21 Reflection Quiz 1&2: CH 5-7

Chapter 5

Electric Charges and Fields



Science off the Sphere: Knitting Needle Experiment

NASA: Johnson Space Center (Houston)

https://www.youtube.com/watch?v=qHrBhgwq__Q

Motion of Charged Particles in An Electric Field

§ —
Lecture 06

Recall Knowledge

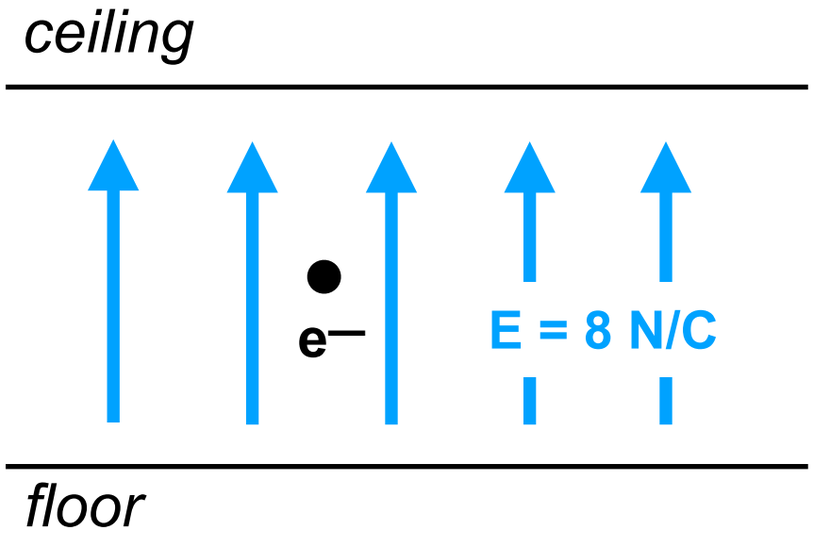
Newton's Laws

Newton's Second Law says…

1. $F_{net} = ma$
2. A force is a push or a pull
3. For each action, there is an equal and opposite reaction
4. $p = m v$
5. Force is a vector
0. Don't wait on my response

Electron In a Room

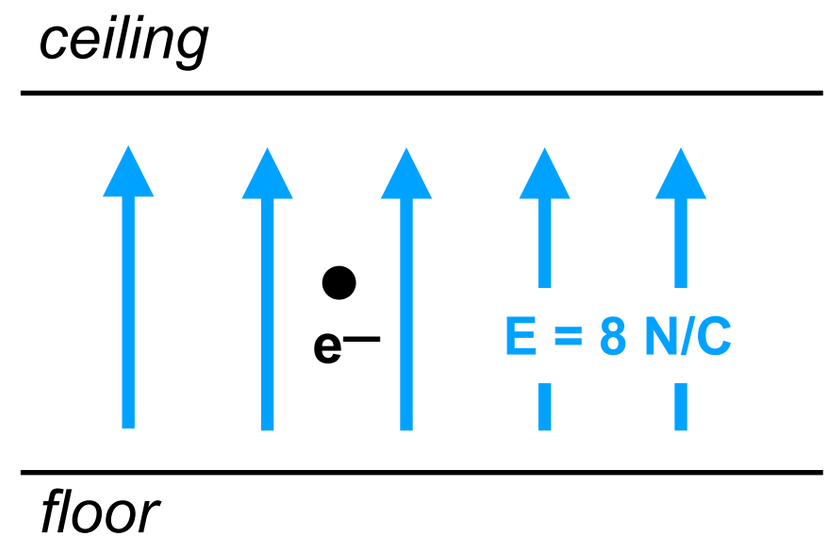
Inside of a 3m tall room, there is a uniform electric field that points directly upward with a magnitude of 8 N/C. An electron is held half-way between the floor and ceiling.



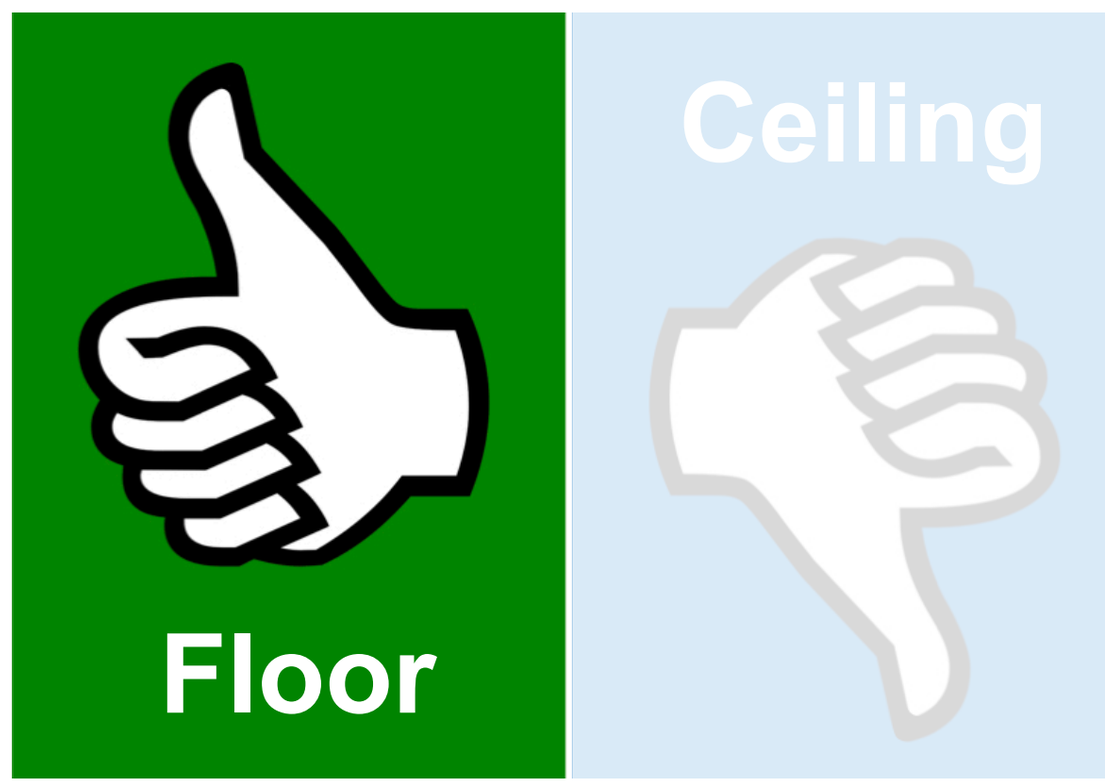
Electron In a Room

Electron motion

Inside of a 3m tall room, there is a uniform electric field that points directly upward with a magnitude of 8 N/C. An electron is held half-way between the floor and ceiling.



After being released, will the electron hit the floor or ceiling?

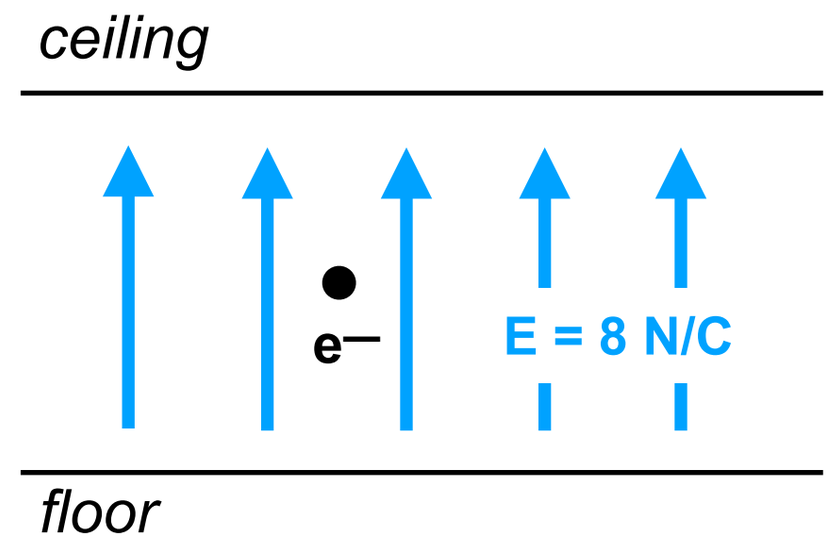


0. Don’t wait on my response

Electron In a Room

Force Comparison

Inside of a 3m tall room, there is a uniform electric field that points directly upward with a magnitude of 8 N/C. An electron is held half-way between the floor and ceiling.



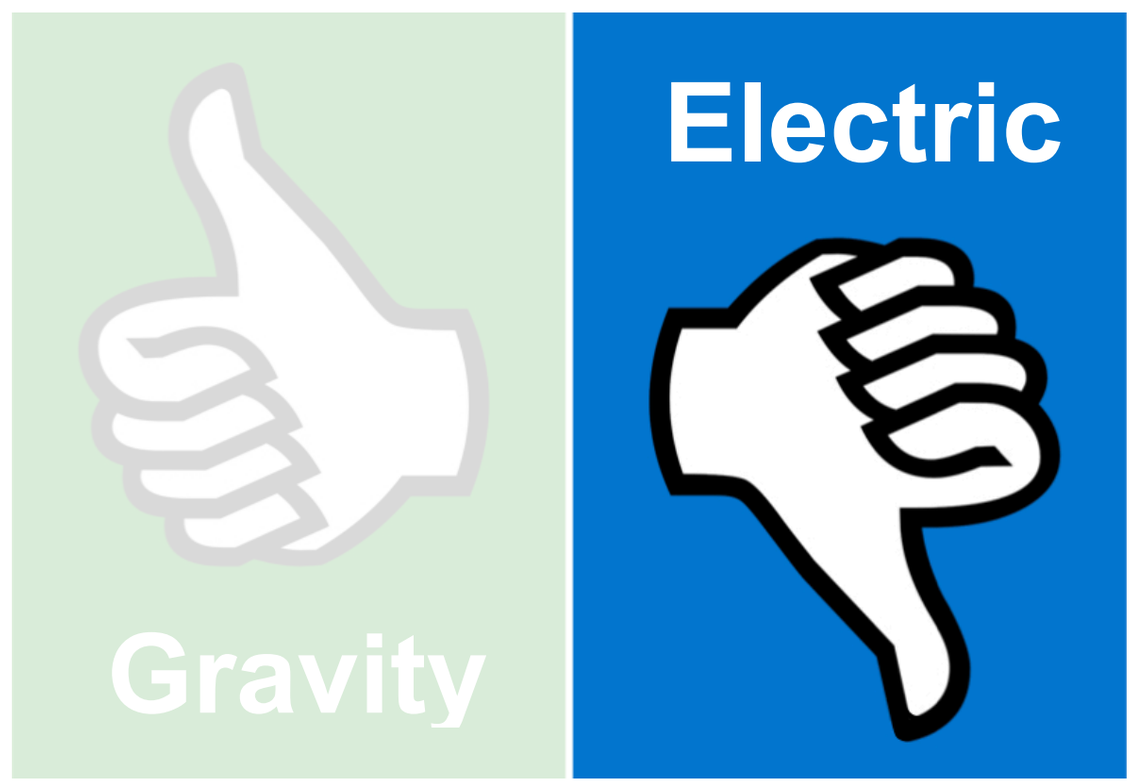
Which force acting on the electron is larger?

$m_e = 9.11 \times 10^{-31}$ kg

$q_e = -1.60 \times 10^{-19}$ C

$F_g = m_e g = 8.9 \times 10^{-30}$ N

$F_e = |q_e|E = 1.3 \times 10^{-18}$ N



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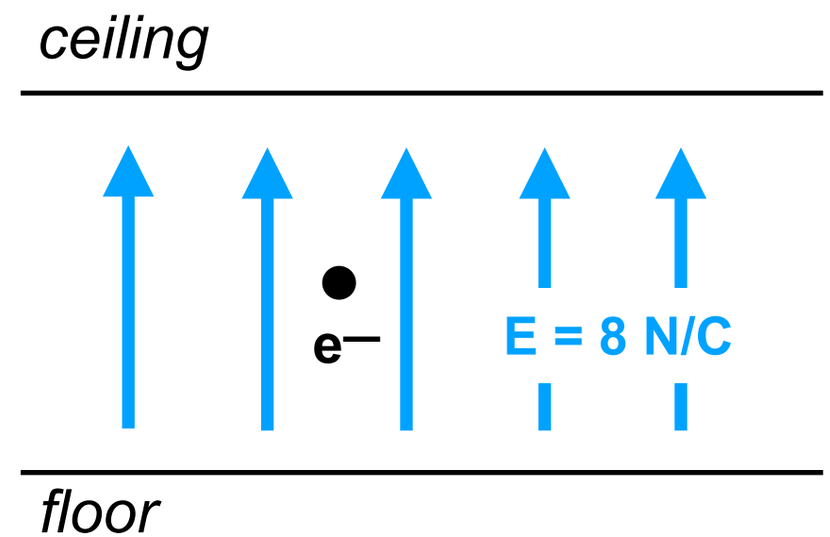
Electron In a Room

Electron's Acceleration

Inside of a 3m tall room, there is a uniform electric field that points directly upward with a magnitude of 8 N/C. An electron is held half-way between the floor and ceiling.

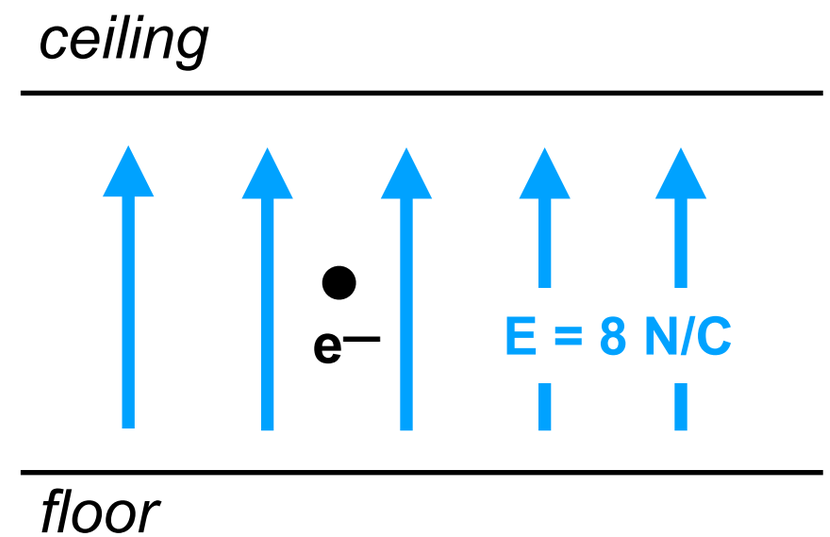


Calculate the acceleration of the electron when it is first released.

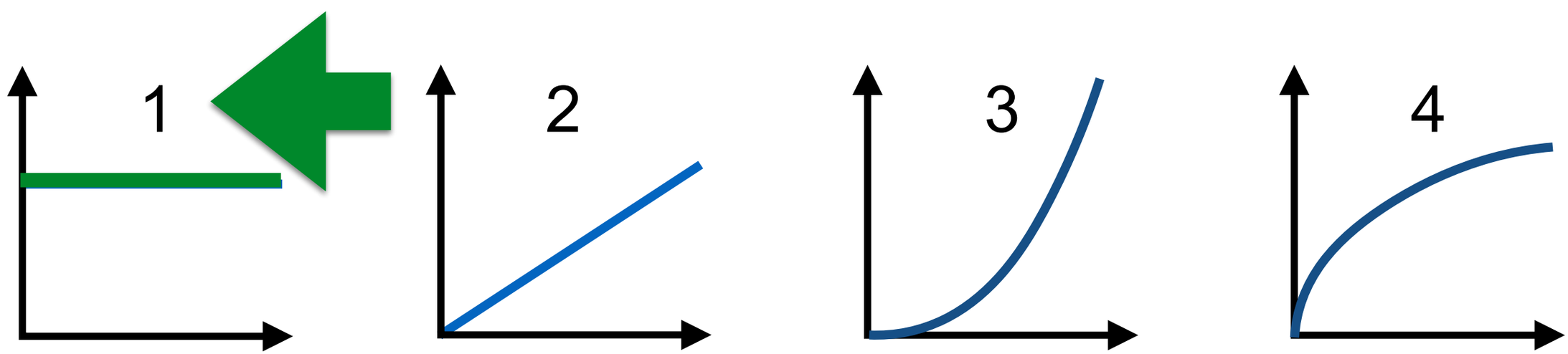
Electron In a Room

Motion Graph - Acceleration

Inside of a 3m tall room, there is a uniform electric field that points directly upward with a magnitude of 8 N/C. An electron is held half-way between the floor and ceiling.



Which graph shows the electron's acceleration (magnitude) over time as it moves?

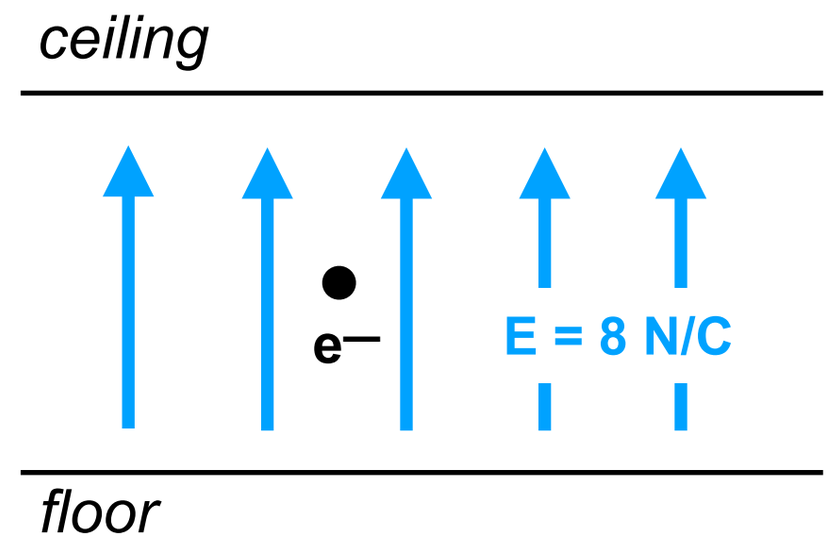


0. Don't wait on my response

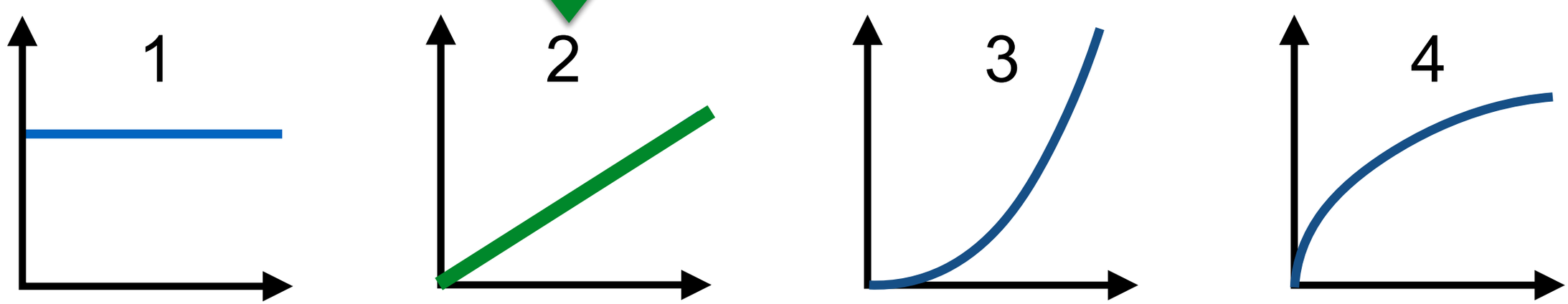
Electron In a Room

Motion Graph - Velocity

Inside of a 3m tall room, there is a uniform electric field that points directly upward with a magnitude of 8 N/C. An electron is held half-way between the floor and ceiling.



Which graph shows the electron's speed over time as it moves?



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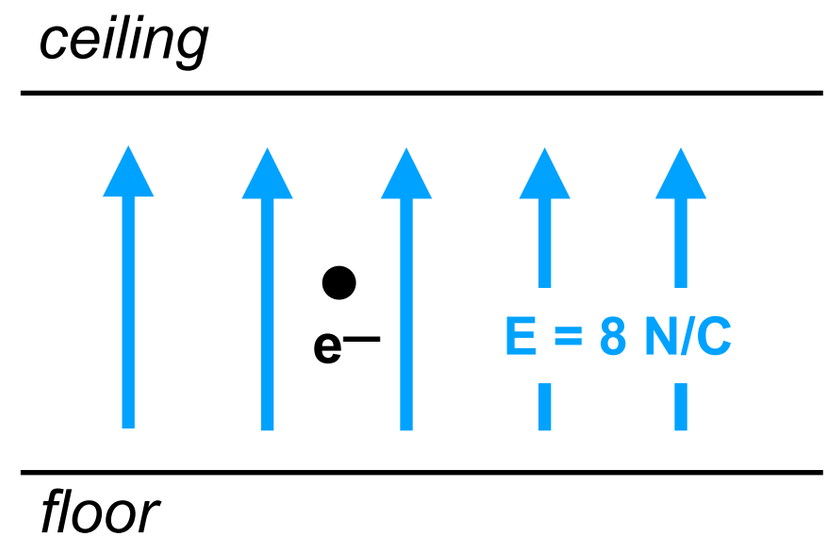
Electron In a Room

Electron's Motion

Inside of a 3m tall room, there is a uniform electric field that points directly upward with a magnitude of 8 N/C. An electron is held half-way between the floor and ceiling.



How long does it take the electron to reach the floor?

How fast is the electron moving when it reaches the floor?

Electrostatic Equilibrium

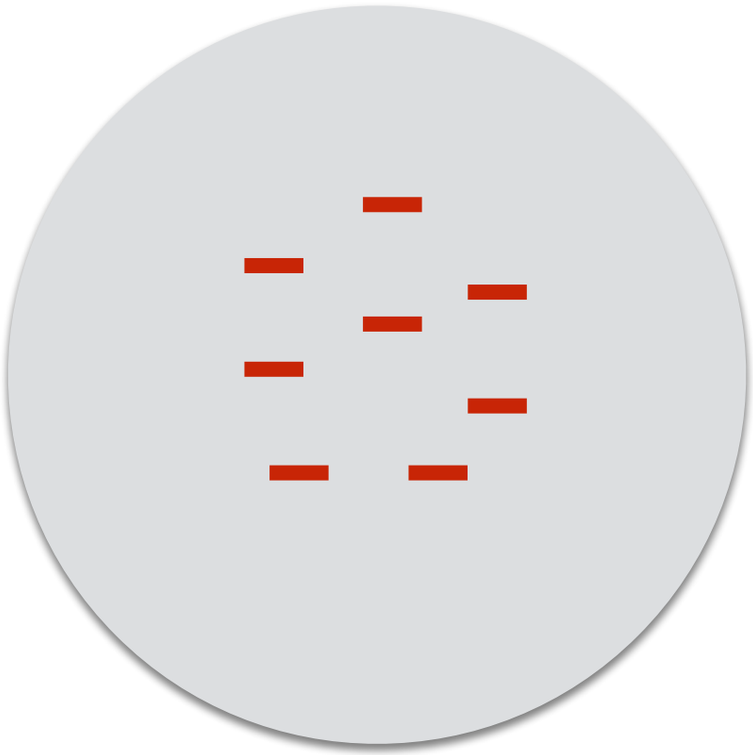
Electric charges feel no net force : $F_{net} = 0$

Inside a conductor, charges are free to move

Inside a <u>charged</u> conductor, charges move until they find electrostatic equilibrium

This happens by charges spreading out on the <u>outer</u> surface of the conductor

Results in E=0 **inside** the conductor



Electrostatic Equilibrium

How?

The charges inside a conductor do not “know” to rearrange until $\vec{E}=0$.

If an electric field appears, they feel a force from that field : $\vec{F} = q\vec{E}$.

As they move around due to that force, the field changes because the charge arrangement changes.

They keep moving (due to the changing field) until they no longer feel any force trying to move them.

When the force is zero, the field is zero : $\vec{E} = \vec{F} / q$.

Chapter 6

Gauss's Law

Electric Flux

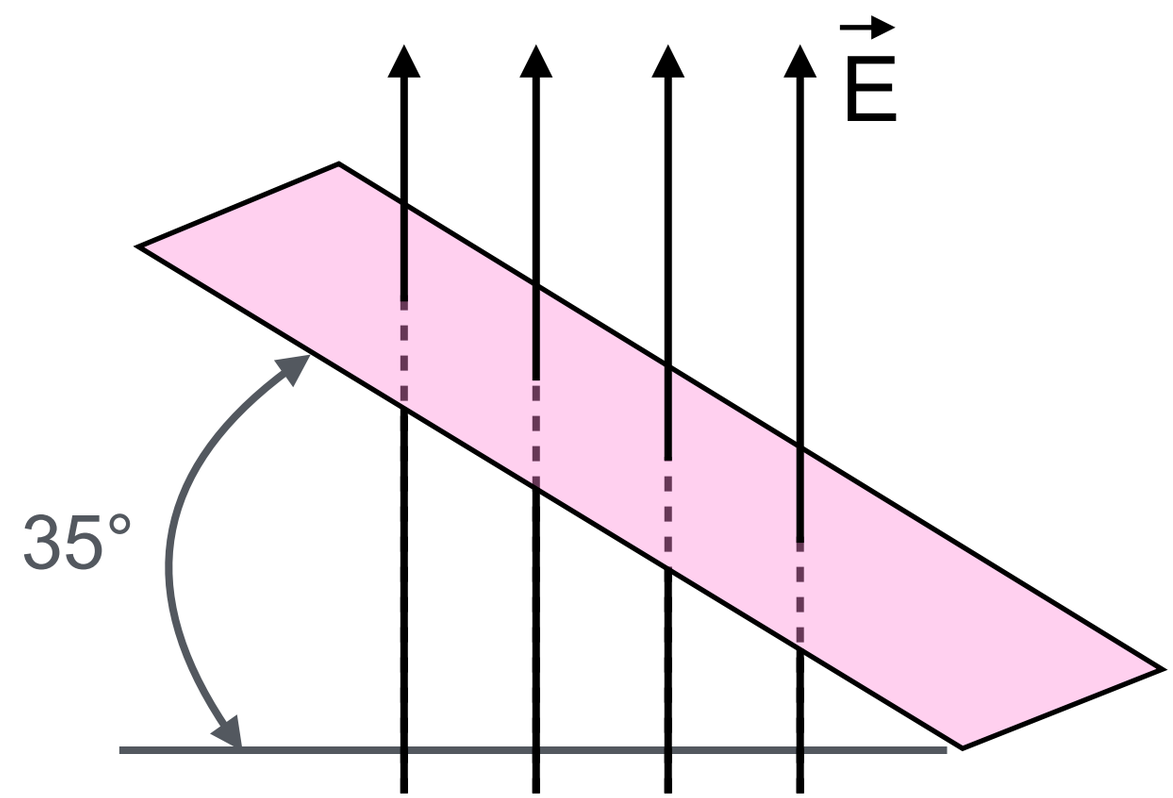
§ 6.1
Lecture 06

Example : Uniform Field

A uniform, vertical electric field with strength 12N/C passes through a 0.25m × 0.10m board, inclined 35° from the horizontal.

Calculate the electric flux through the board.

$$\Phi_E = \int \vec{E} \cdot d\vec{A}$$

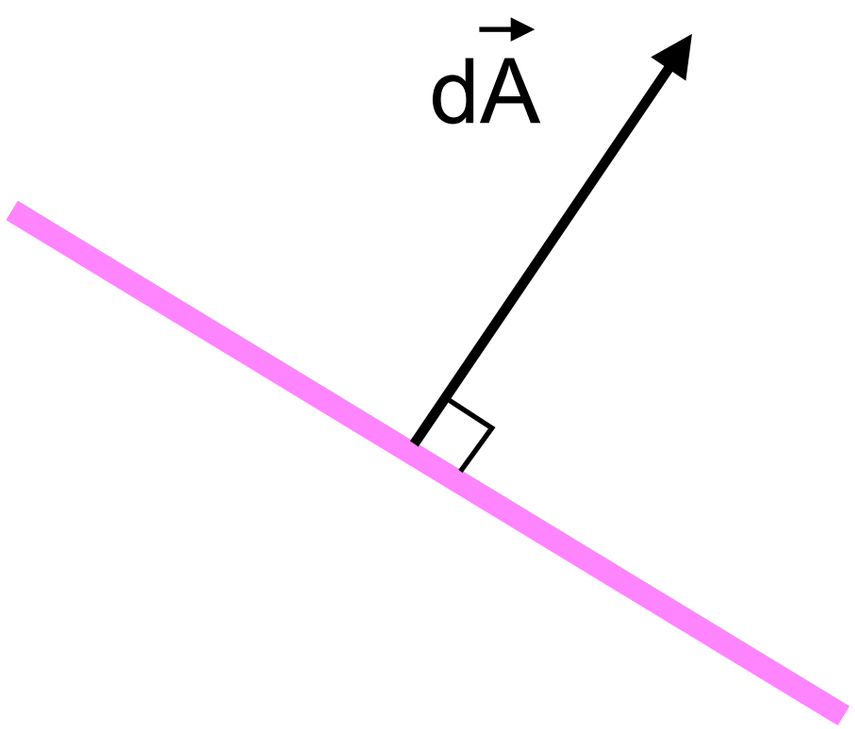


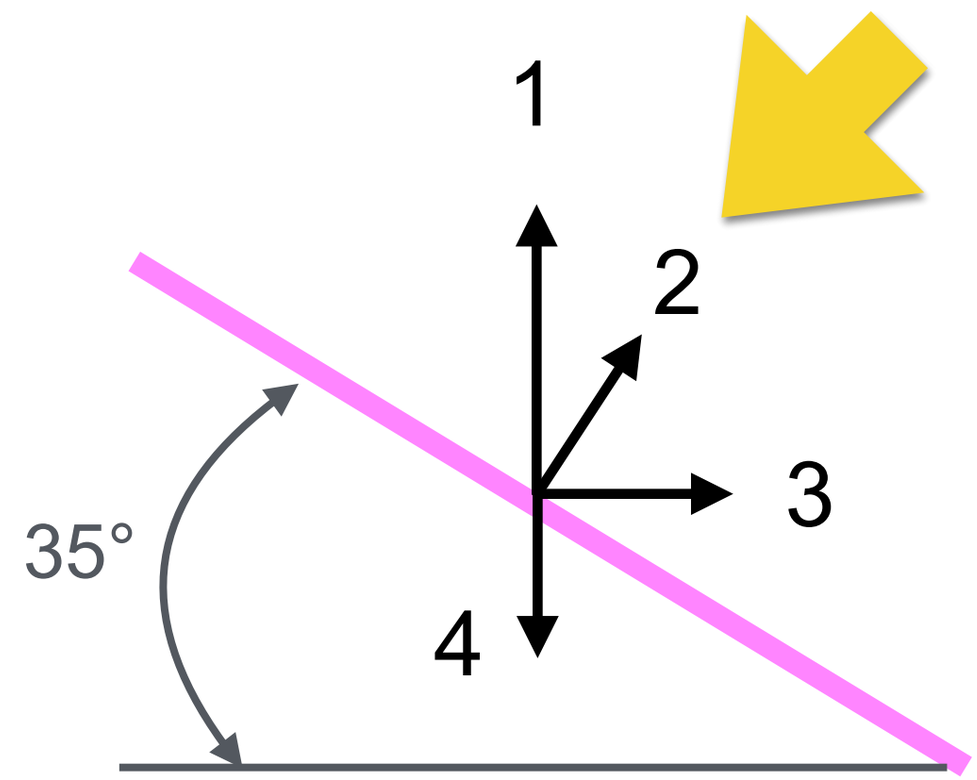
Example : Uniform Field

$d\vec{A}$ Direction

A uniform, vertical electric field with strength 12N/C passes through a 0.25m × 0.10m board, inclined 35° from the horizontal.

What is the direction of $d\vec{A}$?





0. Don’t wait on my response

Example : Uniform Field

Calculation

A uniform, vertical electric field with strength 12N/C passes through a 0.25m × 0.10m board, inclined 35° from the horizontal.

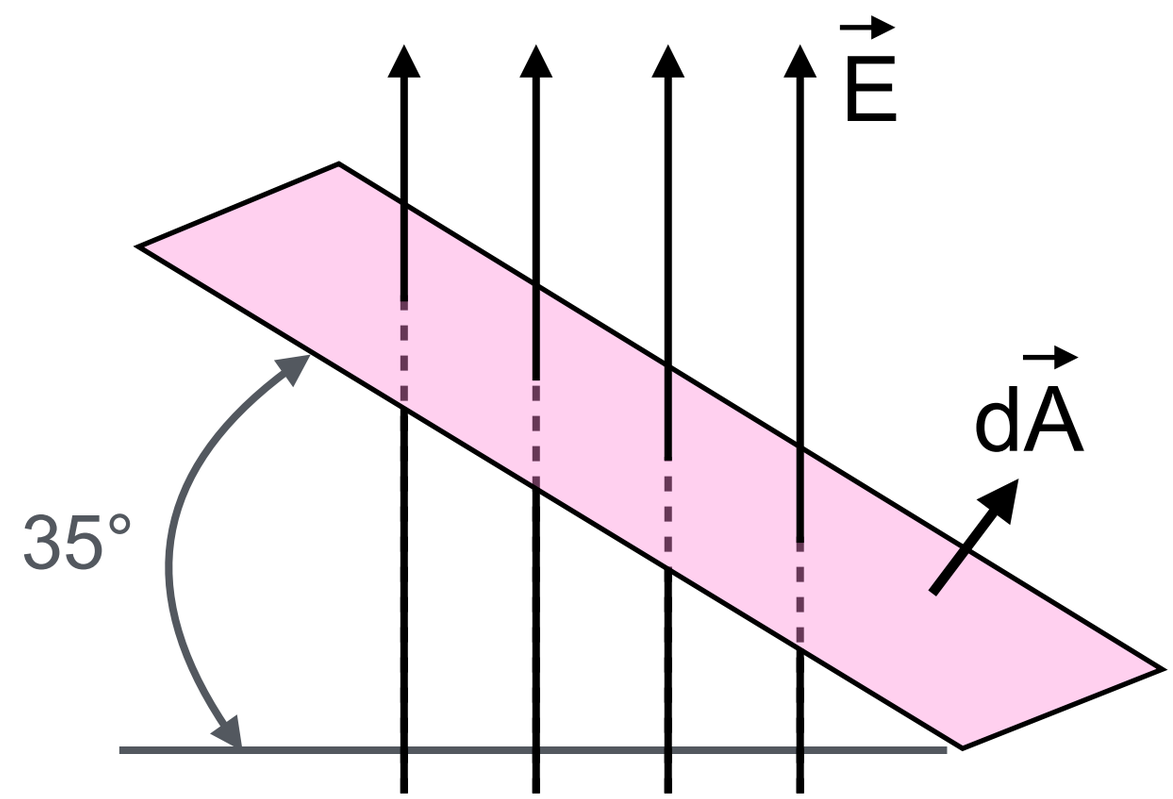
$$\Phi_E = \int \vec{E} \cdot d\vec{A}$$

$$\Phi_E = \int E \, dA \cos\theta$$

$$\Phi_E = E \cos\theta \int dA$$

$$\Phi_E = E \cos\theta \, A$$

$$\Phi_E = E \cos\theta \, (L \times W)$$

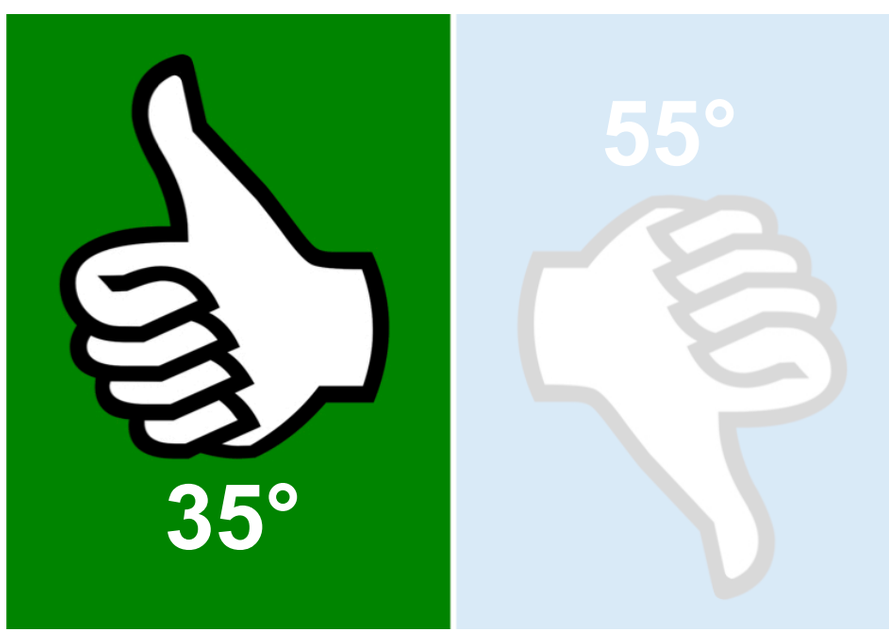


Example : Uniform Field

Vector Angles

A uniform, vertical electric field with strength 12N/C passes through a 0.25m × 0.10m board, inclined 35° from the horizontal.

What is the angle between E and dA?



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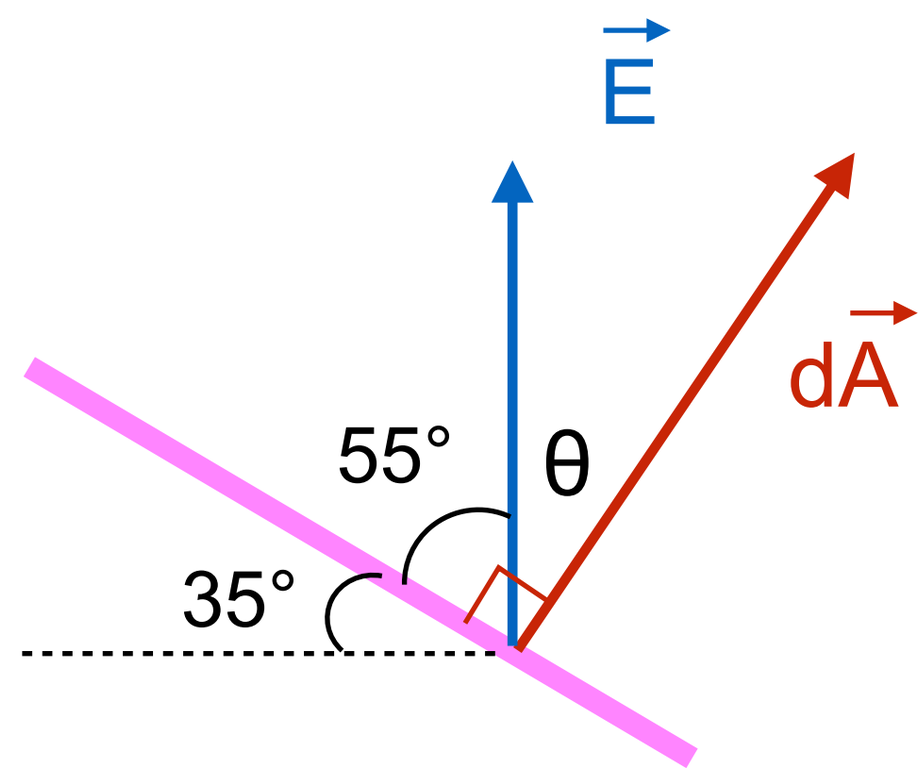


Example : Uniform Field

Numeric Answer

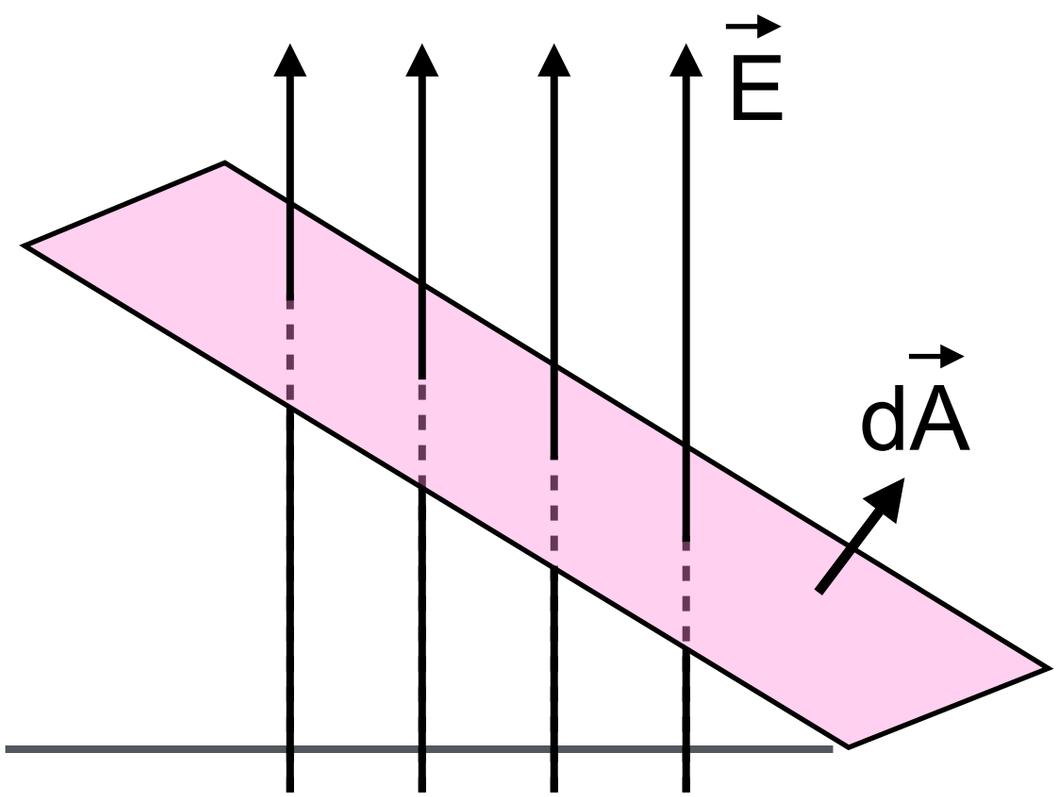
A uniform, vertical electric field with strength 12N/C passes through a 0.25m × 0.10m board, inclined 35° from the horizontal.

Calculate the electric flux through the board.

$\Phi_E = E \cos\theta \, (L \times W)$

$\Phi_E = (12\text{N/C}) \cos 35° \times (0.25\text{m} \times 0.10\text{m})$

$\Phi_E = 0.25 \text{ Nm}^2/\text{C}$



Example : Uniform Field

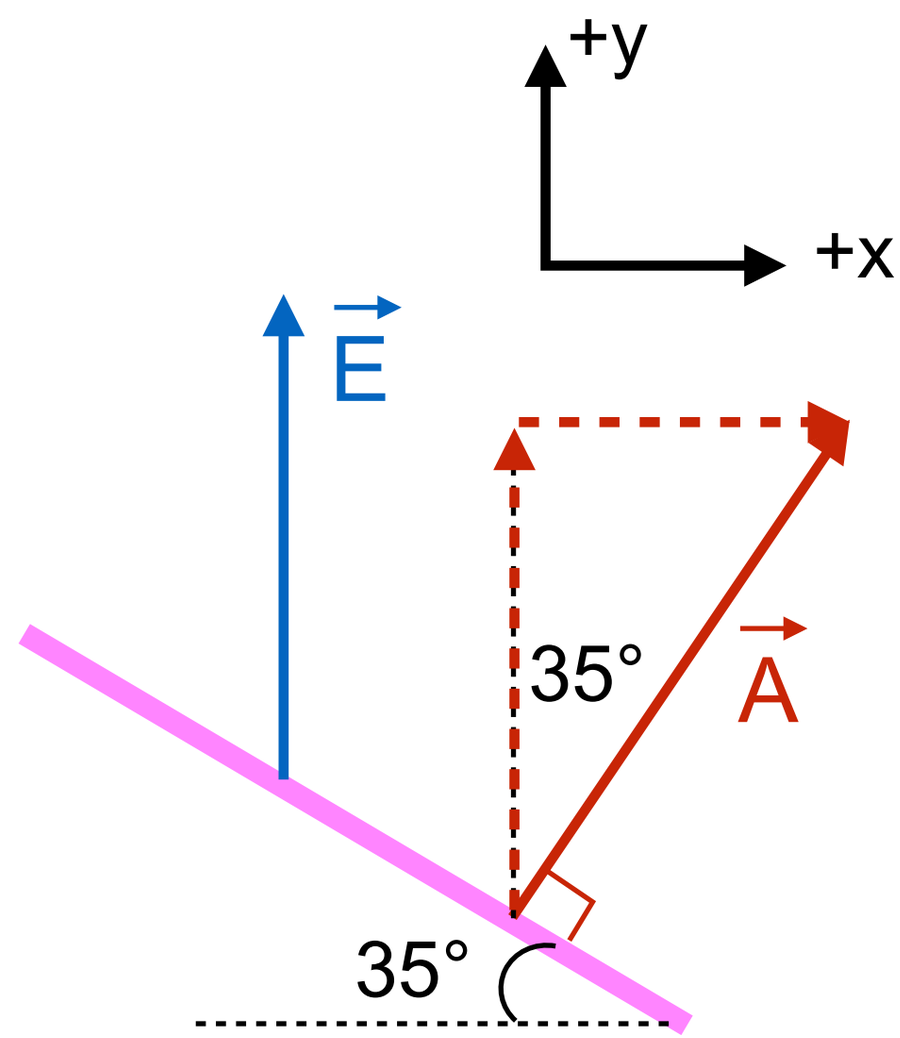
Alternate Solution

A uniform, vertical electric field with strength 12N/C passes through a 0.25m × 0.10m board, inclined 35° from the horizontal.

Write E and A as *vectors*

$\vec{E} = 12\ \text{N/C}\ \hat{j}$

$\vec{A} = (0.025\text{m}^2)\sin(35°)\,\hat{i}$
$\quad + (0.025\text{m}^2)\cos(35°)\,\hat{j}$



Dot product:

$\vec{E} \bullet \vec{A} = E_x A_x + E_y A_y + E_z A_z$

$= (0)(0.025\sin(35°)) +$
$(12)(0.025\cos(35°)) +$
$(0)(0) \qquad = 0.25\ \text{Nm}^2/\text{C}$

Pre-class Slides

Electric Flux

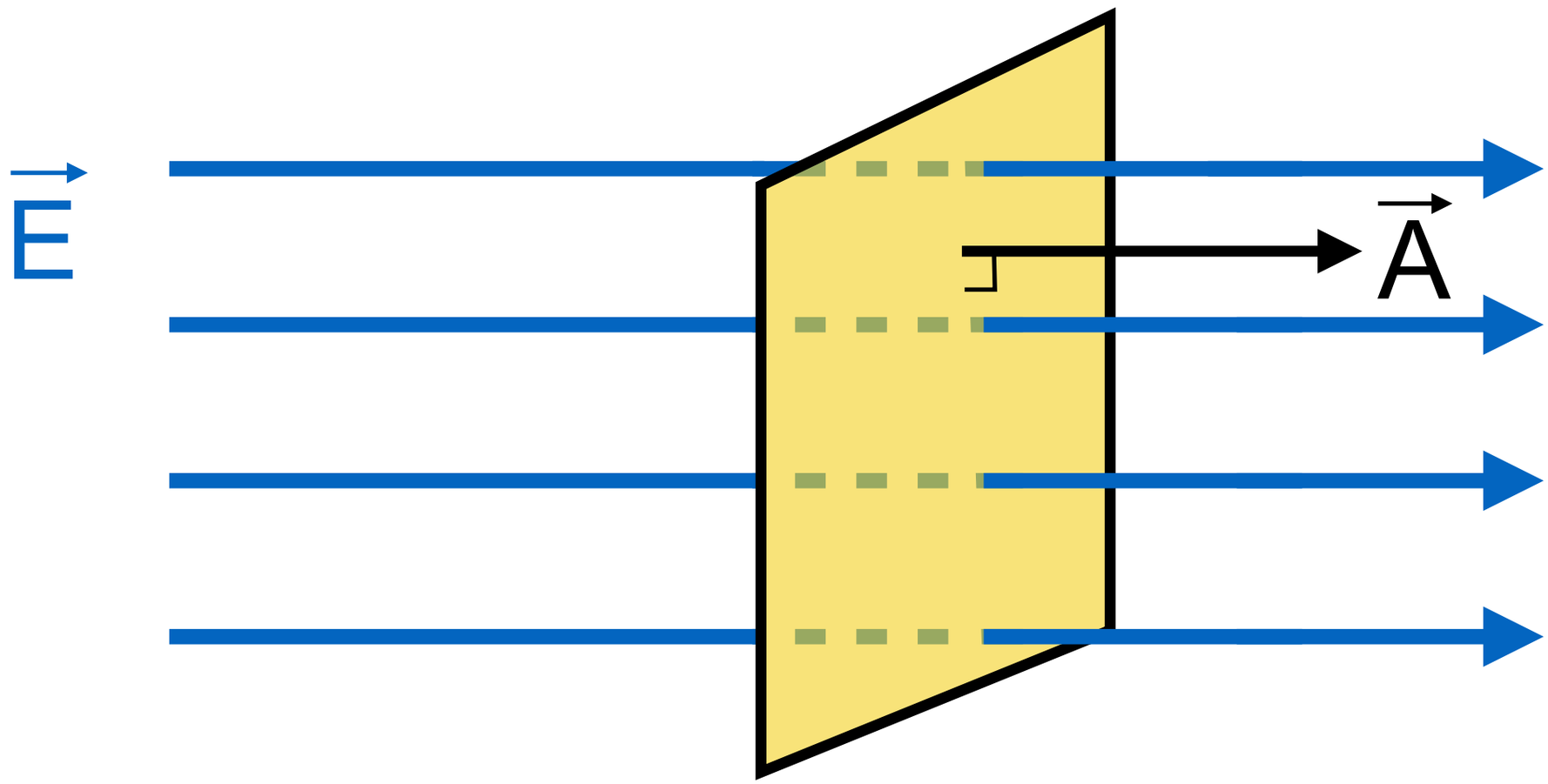
§ 6.1
Lecture 06

Youtube Playlist

Electric Flux

Amount of electric field passing **through** a surface.

$$\Phi_E = \vec{E} \cdot \vec{A}$$



$\vec{A}$ - magnitude is area of surface

$\vec{A}$ - direction is perpendicular to surface

Flux Units

What are the SI units for electric flux?

1. m^2
2. $N\ C\ m^2$
3. J / m
4. $N\ m^2 / C$
5. $C^2\ m^2$
0. Don't wait on my response

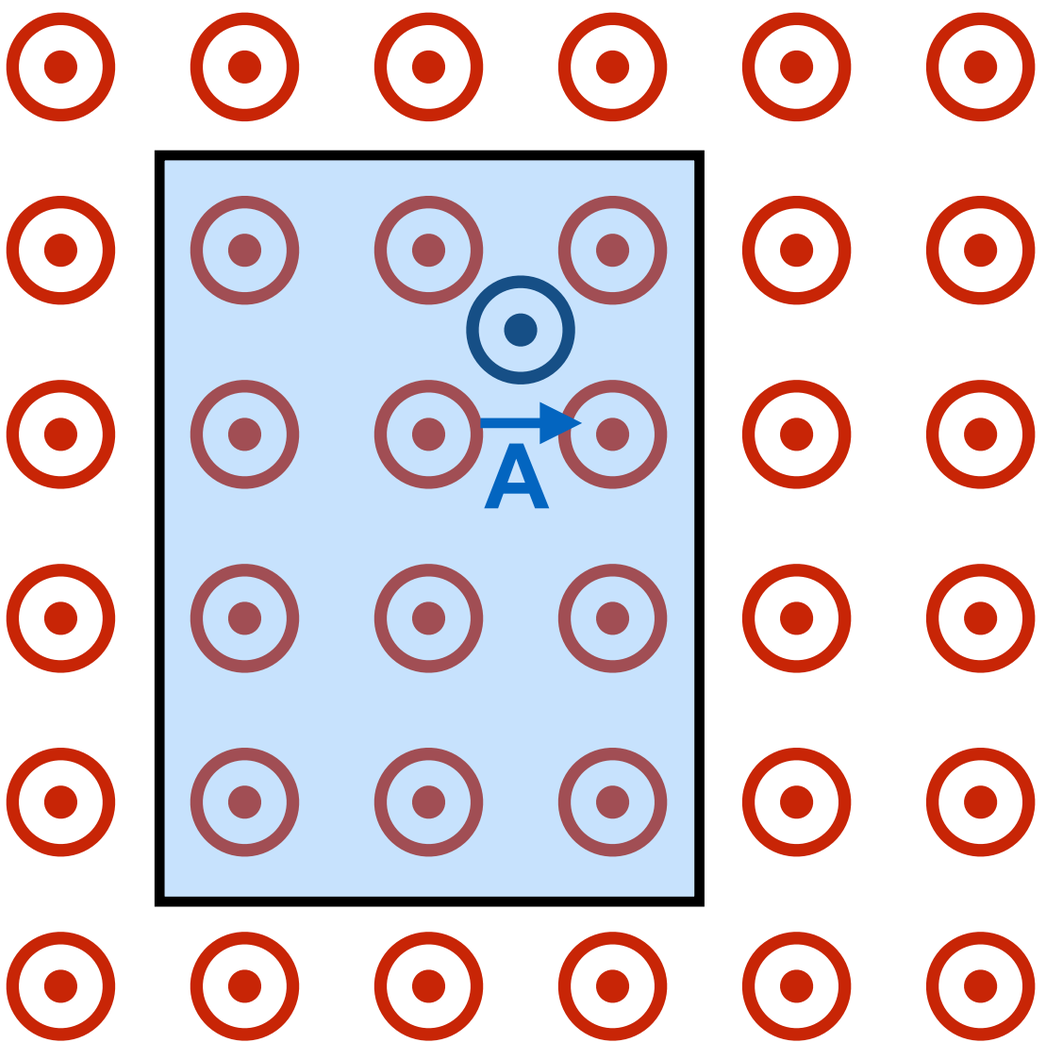
$$\Phi_E = \vec{E} \cdot \vec{A}$$

$$\frac{N}{C} \qquad m^2$$

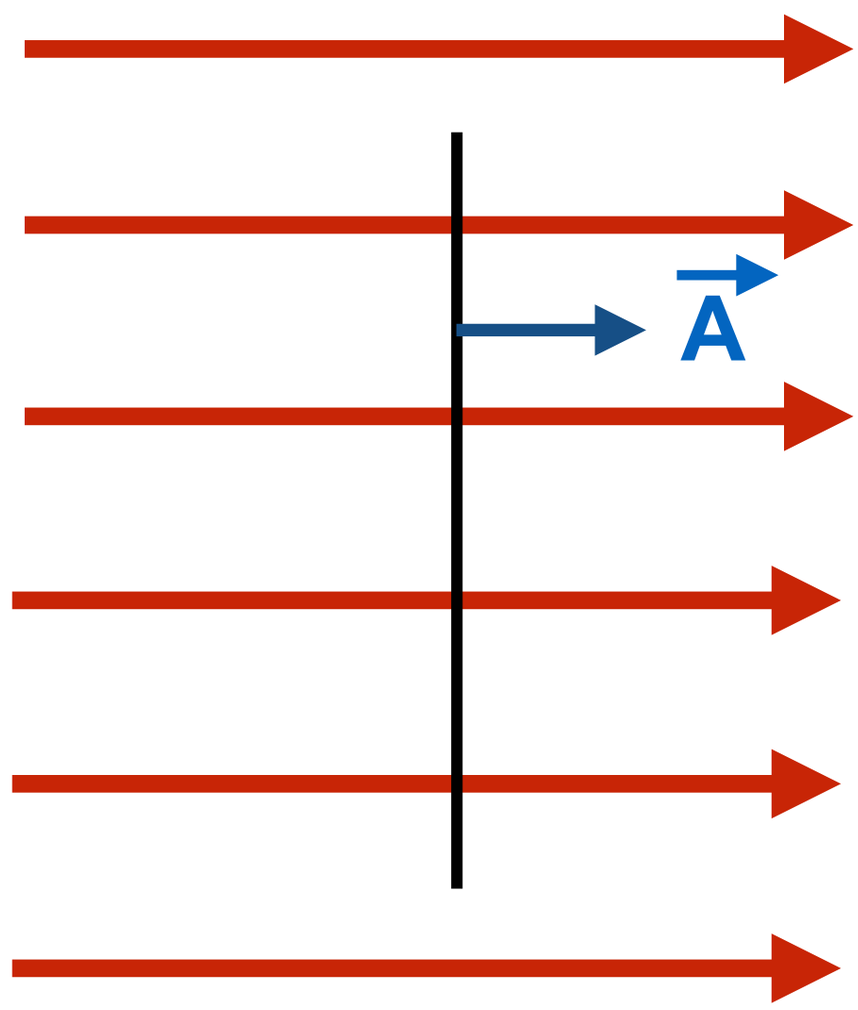
Visualizing Flux

A uniform, constant electric field is horizontal.

Front view



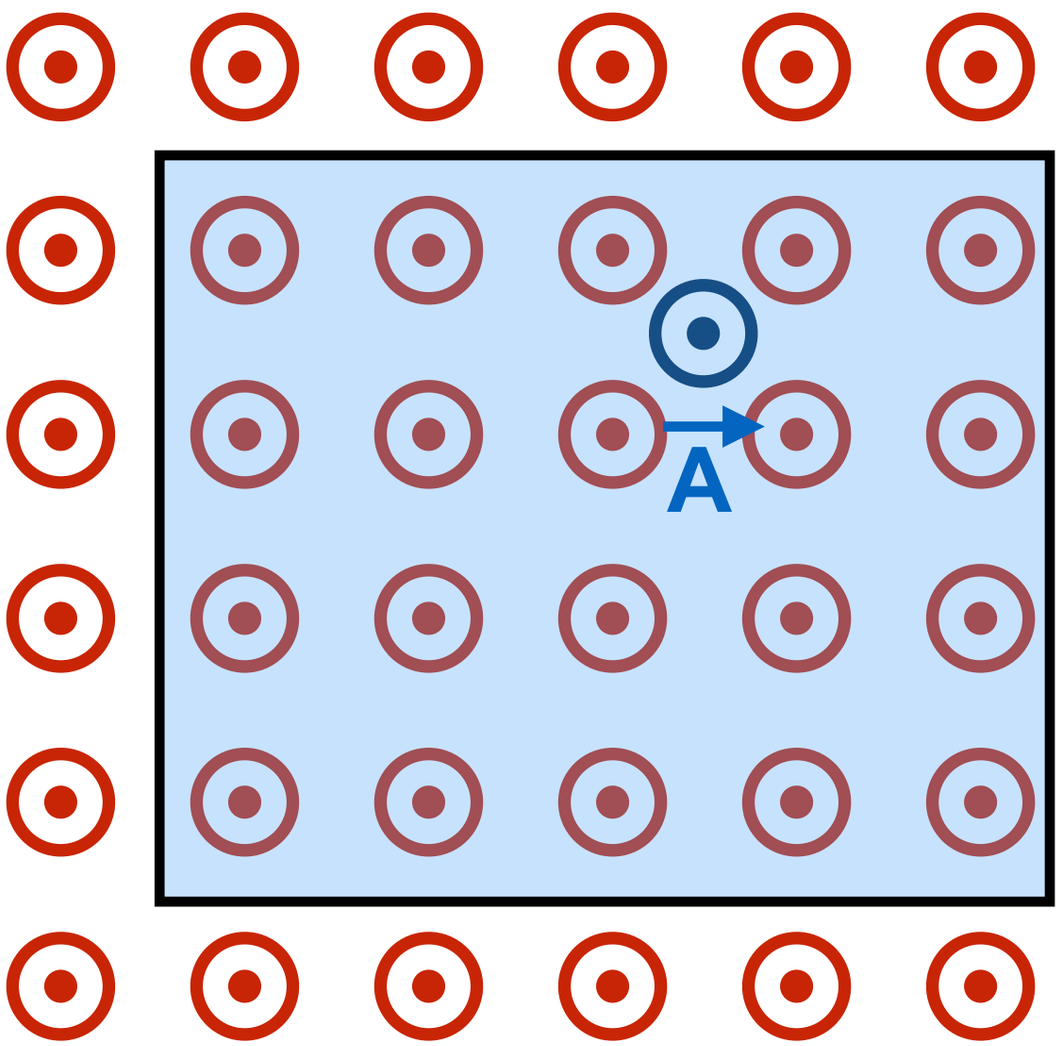
Side View



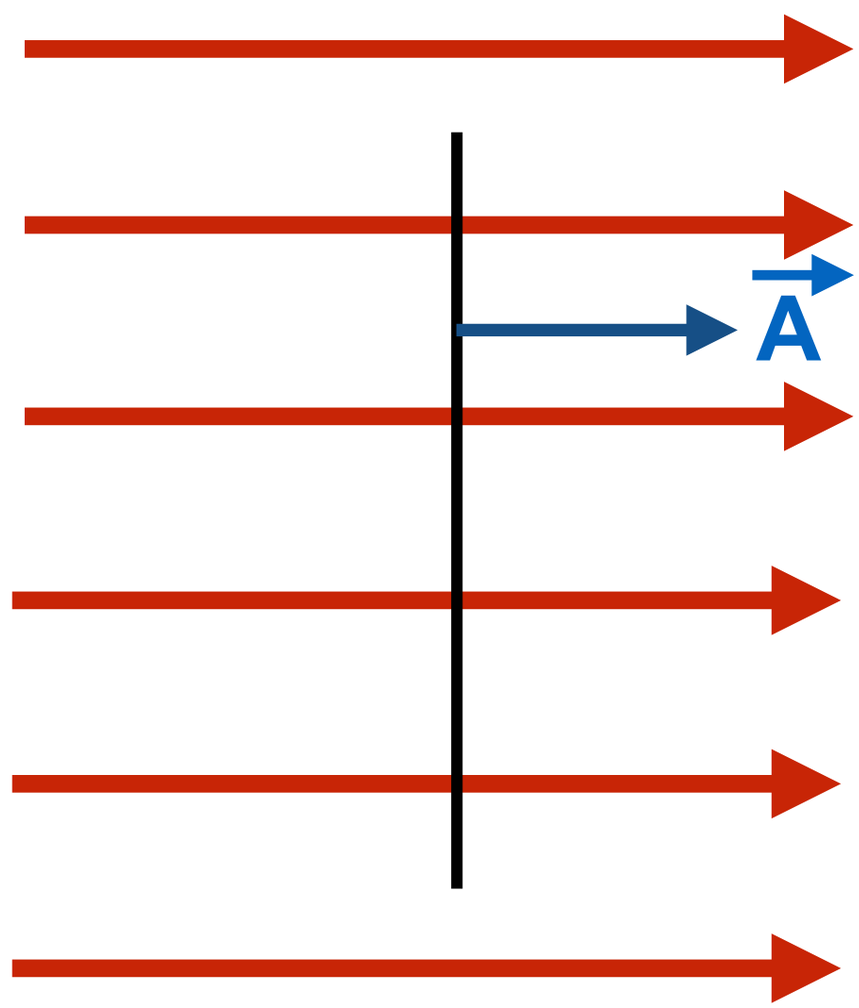
Visualizing Flux

A uniform, constant electric field is horizontal.

Front view



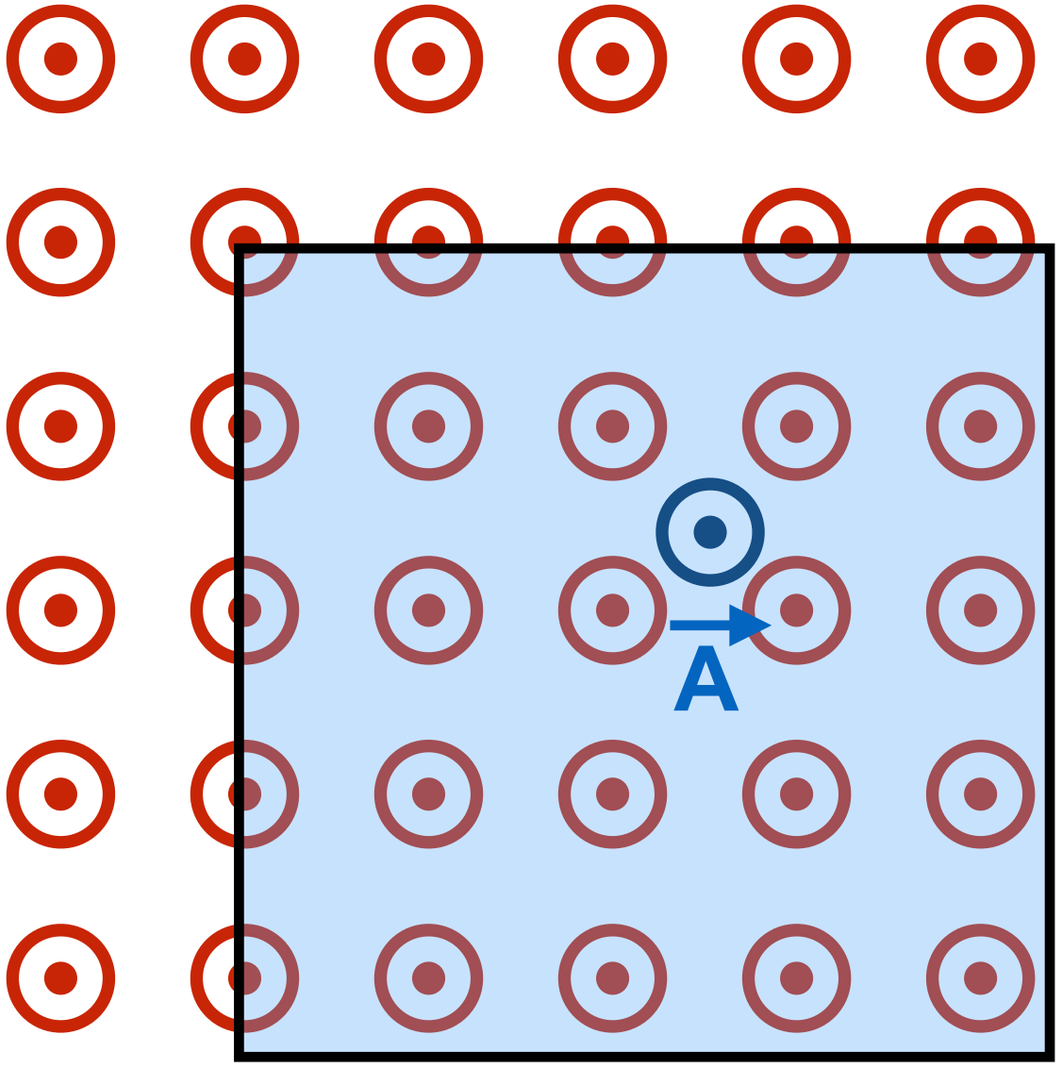
Side View



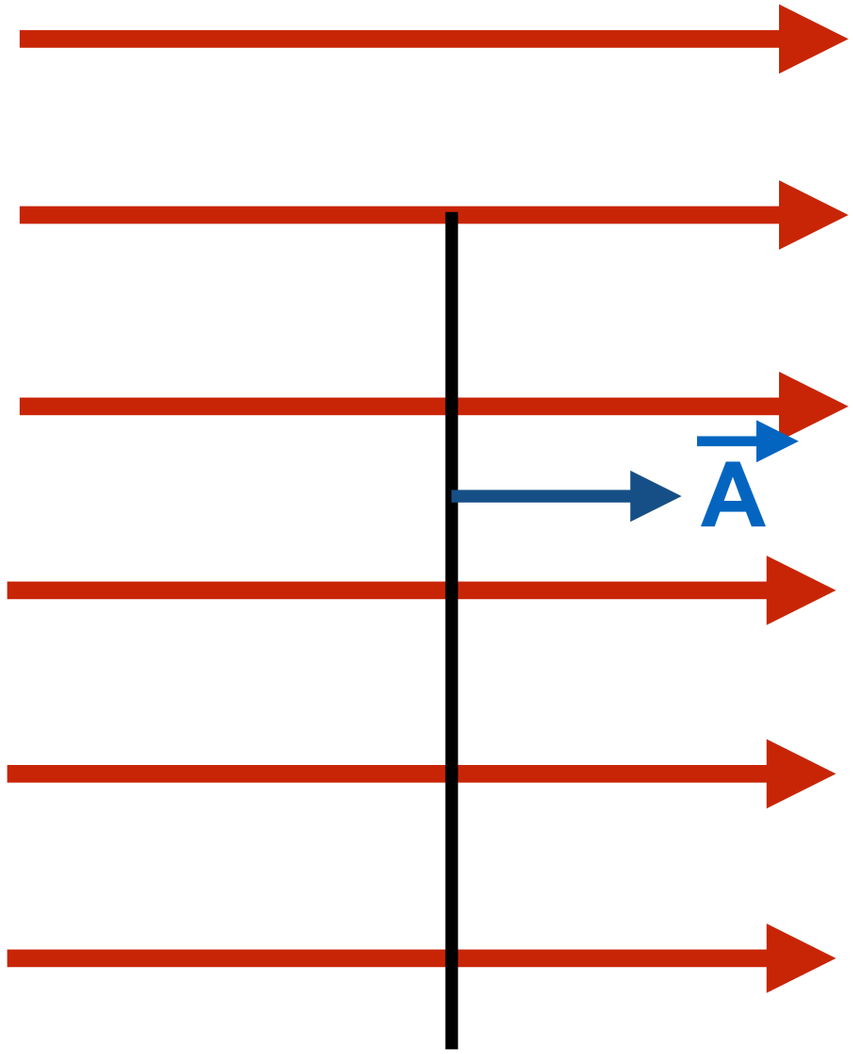
Visualizing Flux

A uniform, constant electric field is horizontal.

Front view



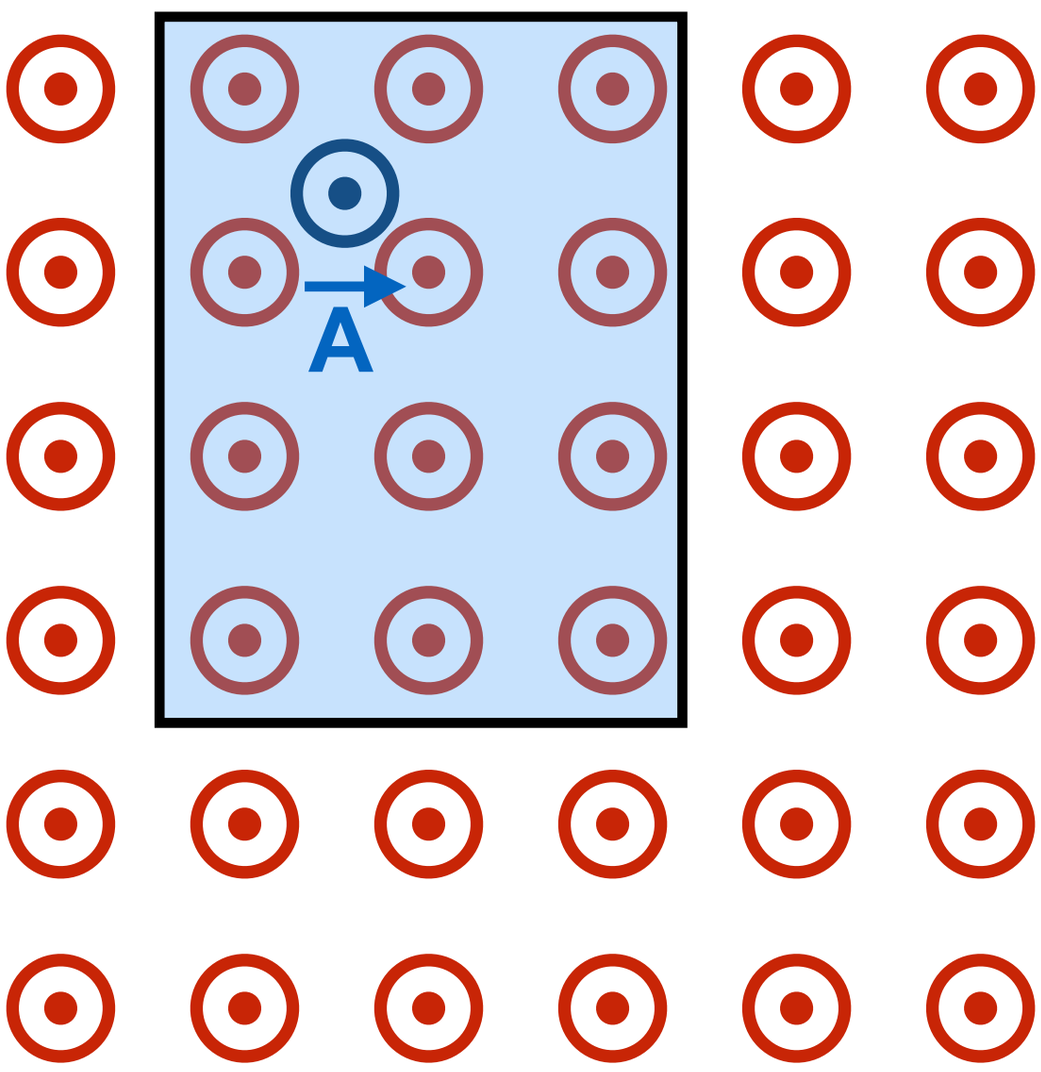
Side View



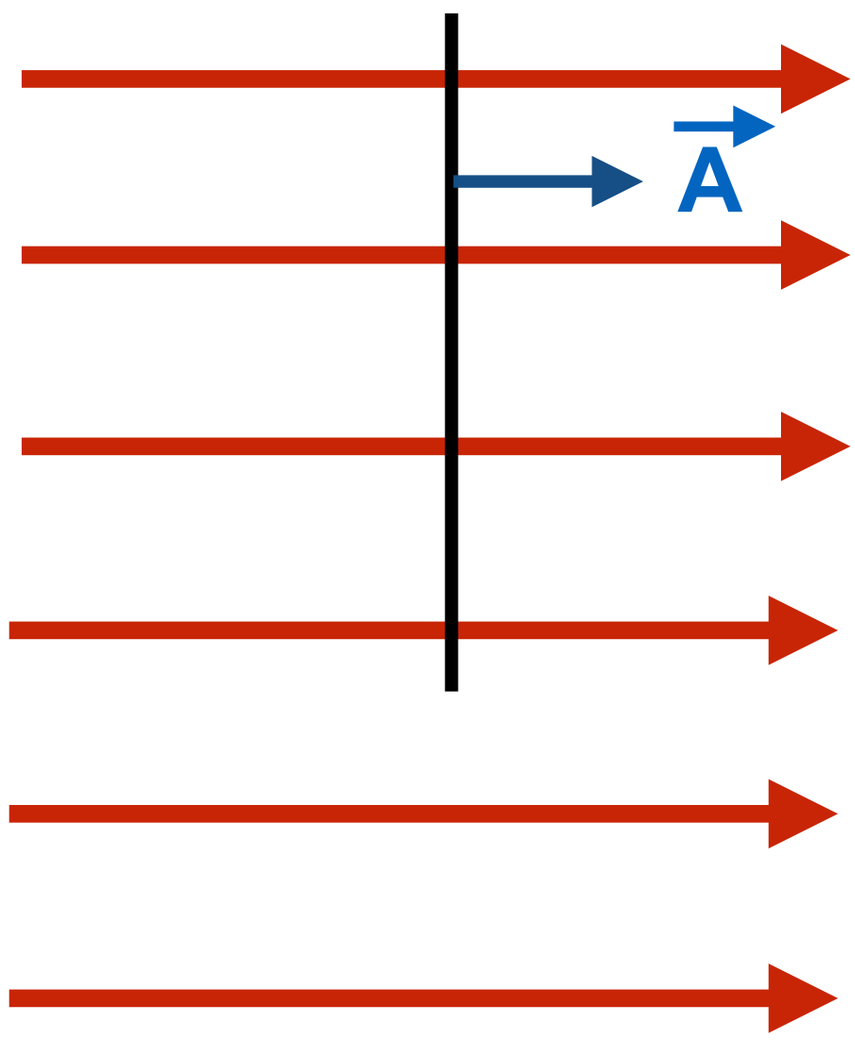
Visualizing Flux

A uniform, constant electric field is horizontal.

Front view



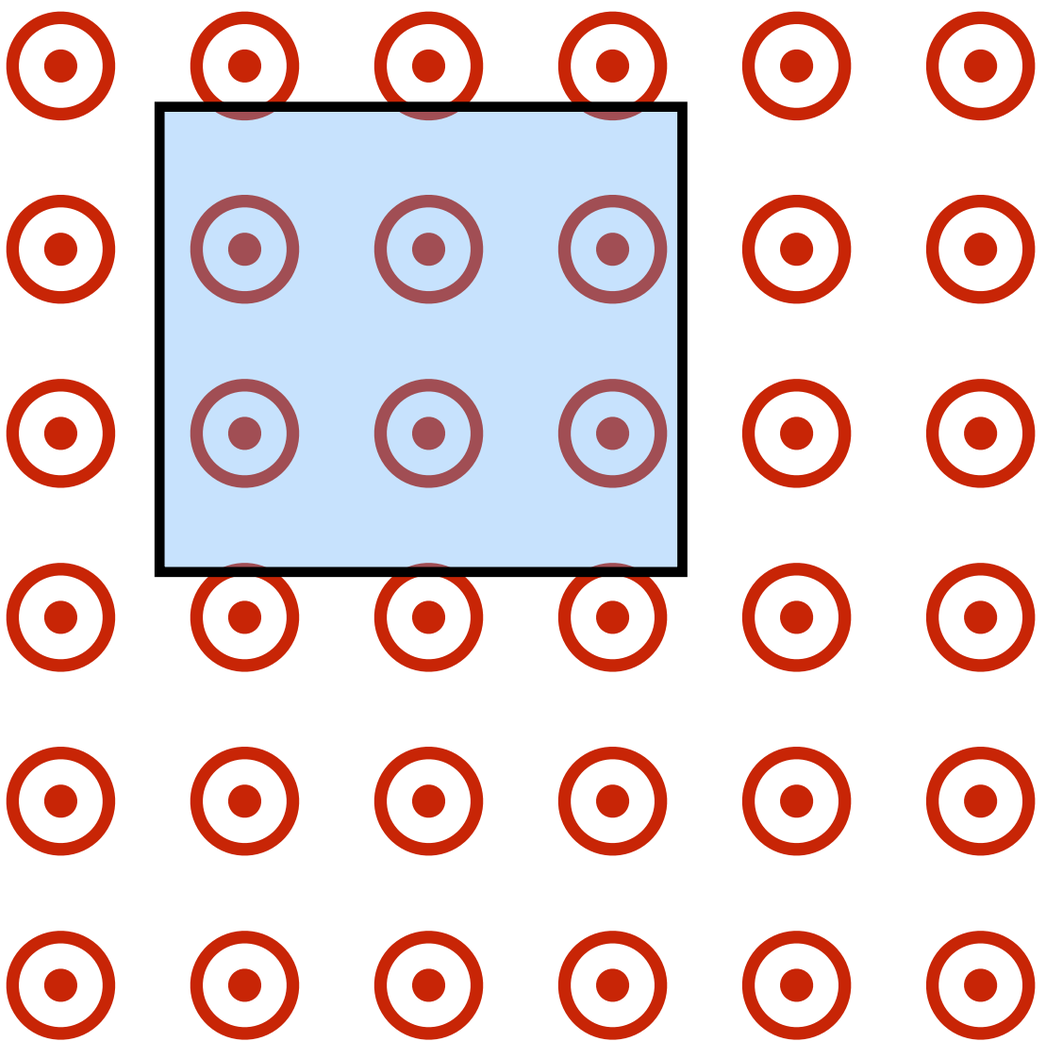
Side View



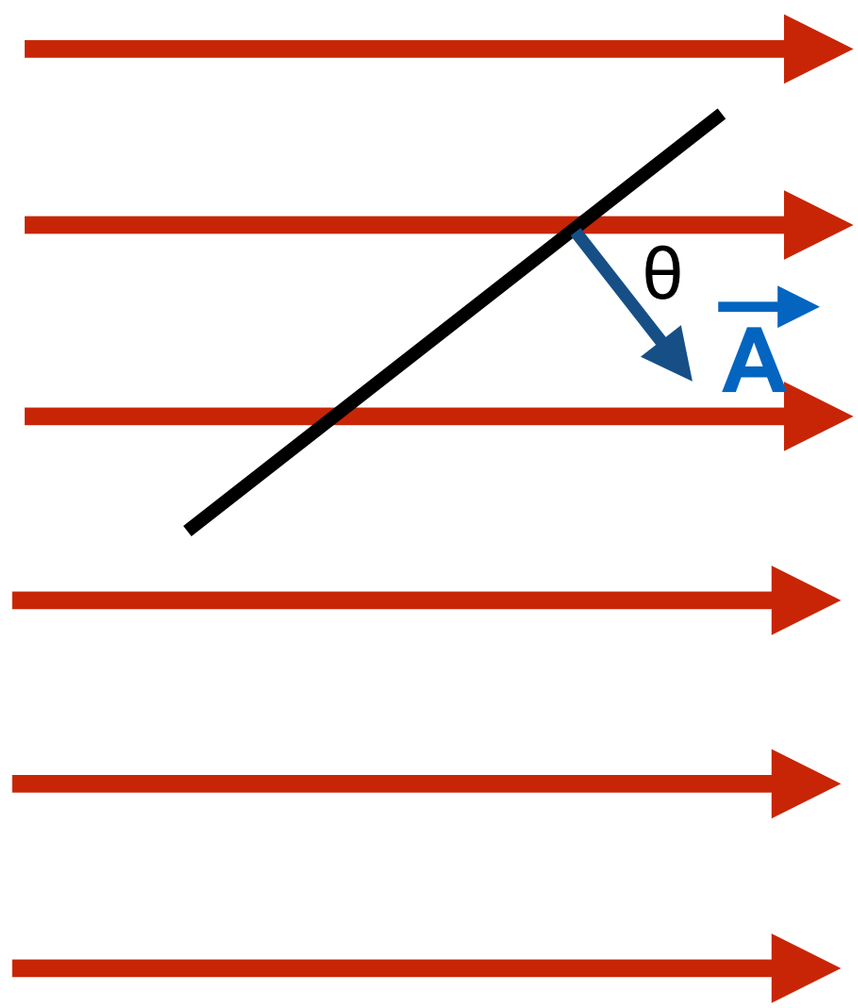
Visualizing Flux

A uniform, constant electric field is horizontal.

Front view



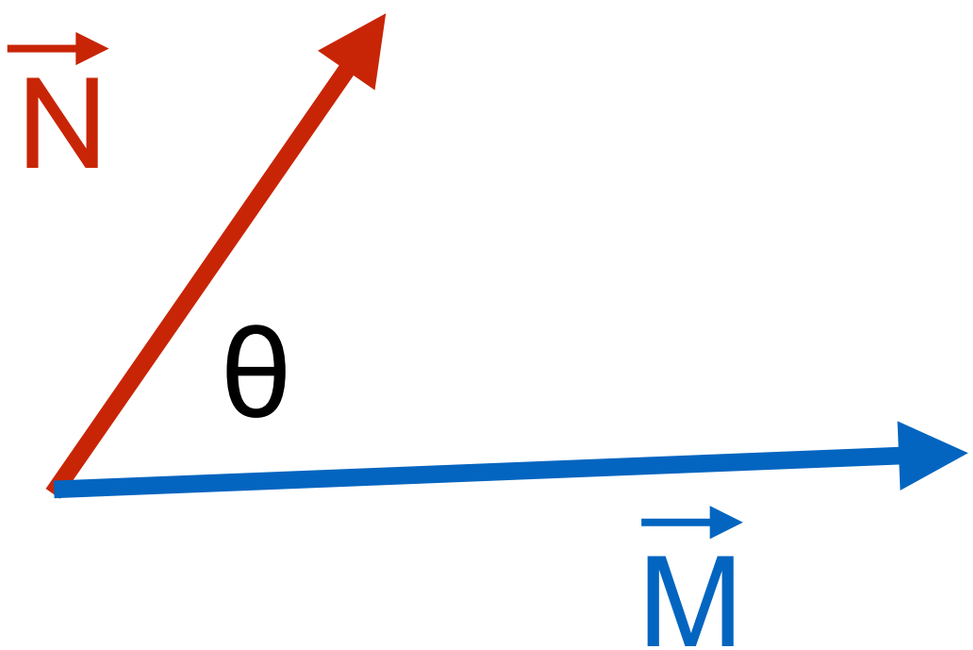
Side View



The Dot Product

Review

Consider the two vectors $\vec{N}$ and $\vec{M}$ separated by an angle θ.



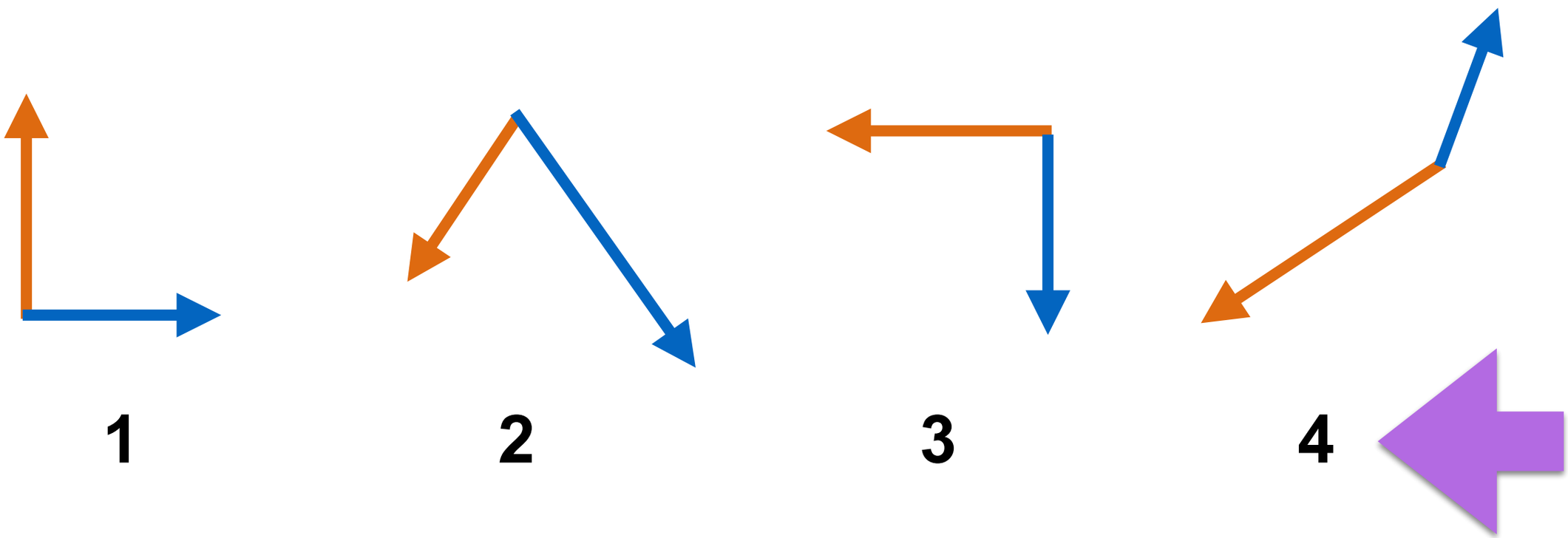
$$\vec{N} \bullet \vec{M} = |N|\,|M| \cos\theta$$

$$\vec{N} \bullet \vec{M} = N_x M_x + N_y M_y + N_z M_z$$

Result is a *scalar*.

Dot Products & Sign

The dot product between which of these vector pairs is negative?

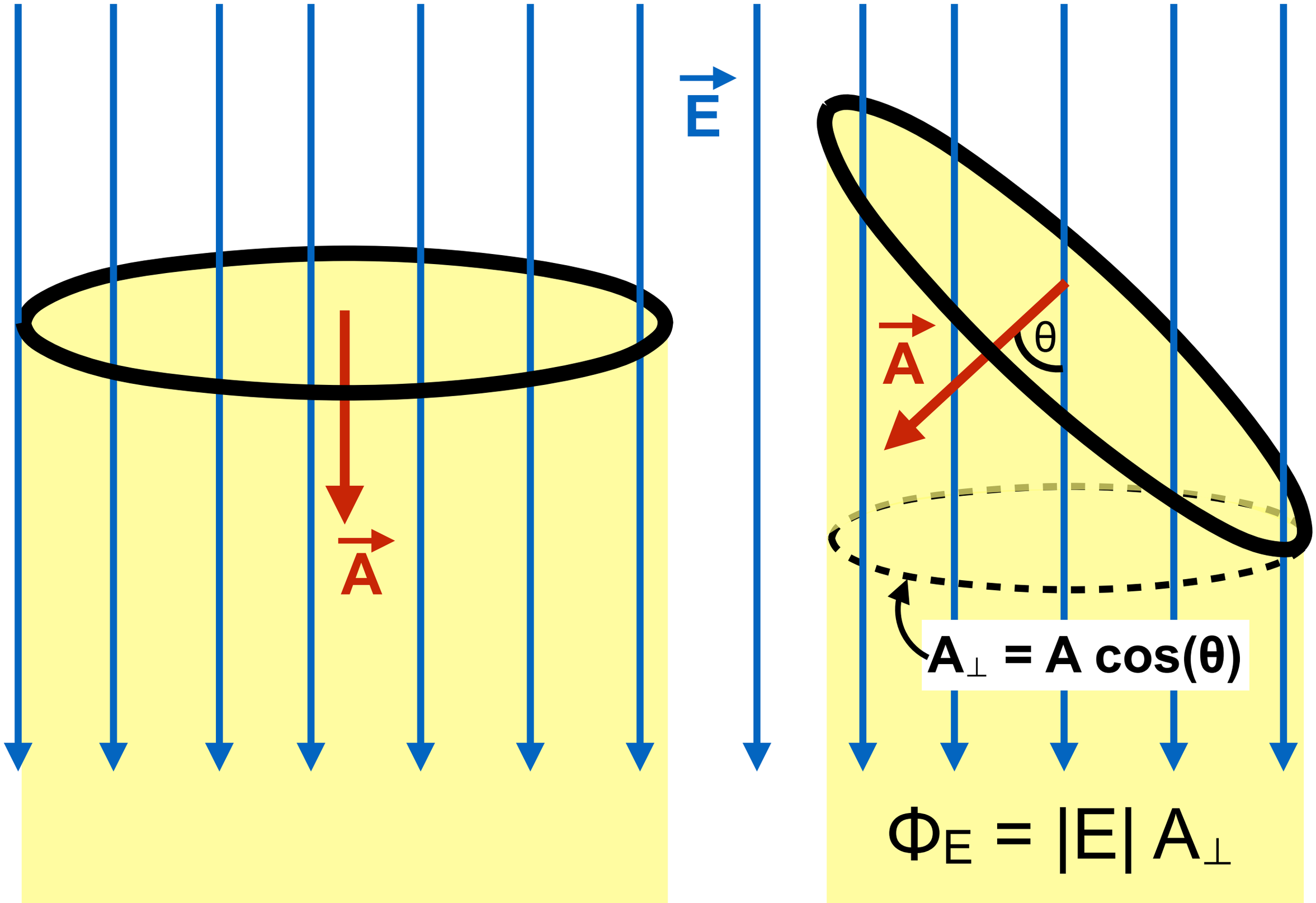


0. Don’t wait on my response

$\cos(\theta)$ is negative if $\theta > 90°$

Visualizing Flux

Rain Analogy



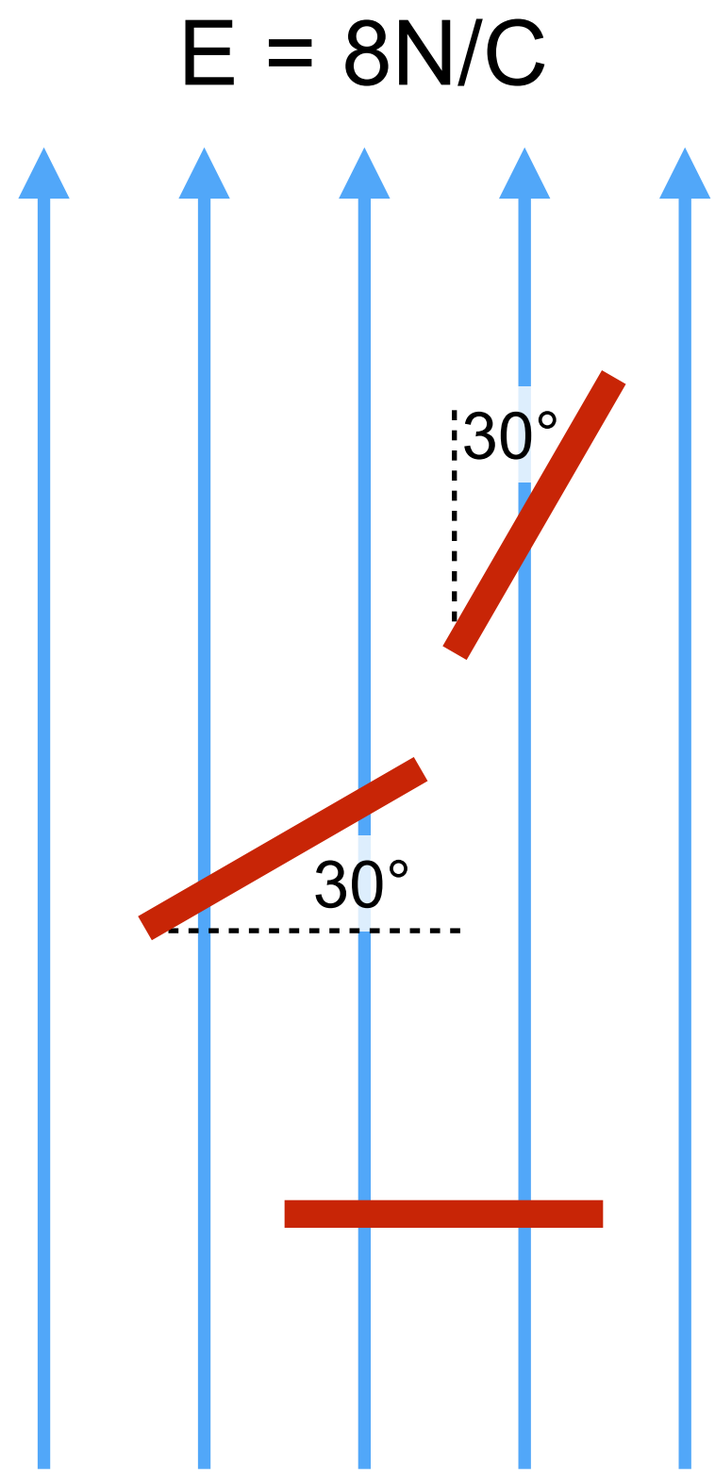
Example : Vertical Field

A uniform electric field points straight upwards with strength 8N/C.

Three circles with radius 0.25m are in the field, oriented as shown.

Which has the **smallest** flux?

1. Top
2. Middle
3. Bottom
4. All the same
0. Don’t wait on my response



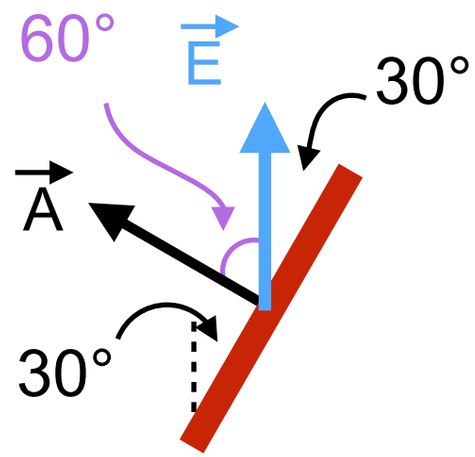
Example : Vertical Field

Top : $\Phi_E = \vec{E} \cdot \vec{A}$

$= E\,A\cos(60°)$

$= E\,\pi\,R^2\cos(60°)$

$= 0.78\ \mathrm{Nm^2/C}$

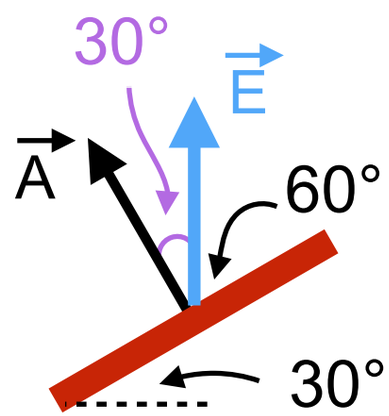


Middle : $\Phi_E = \vec{E} \cdot \vec{A}$

$= E\,A\cos(30°)$

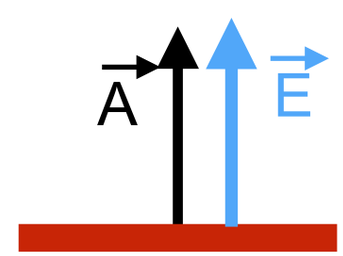
$= E\,\pi\,R^2\cos(30°)$

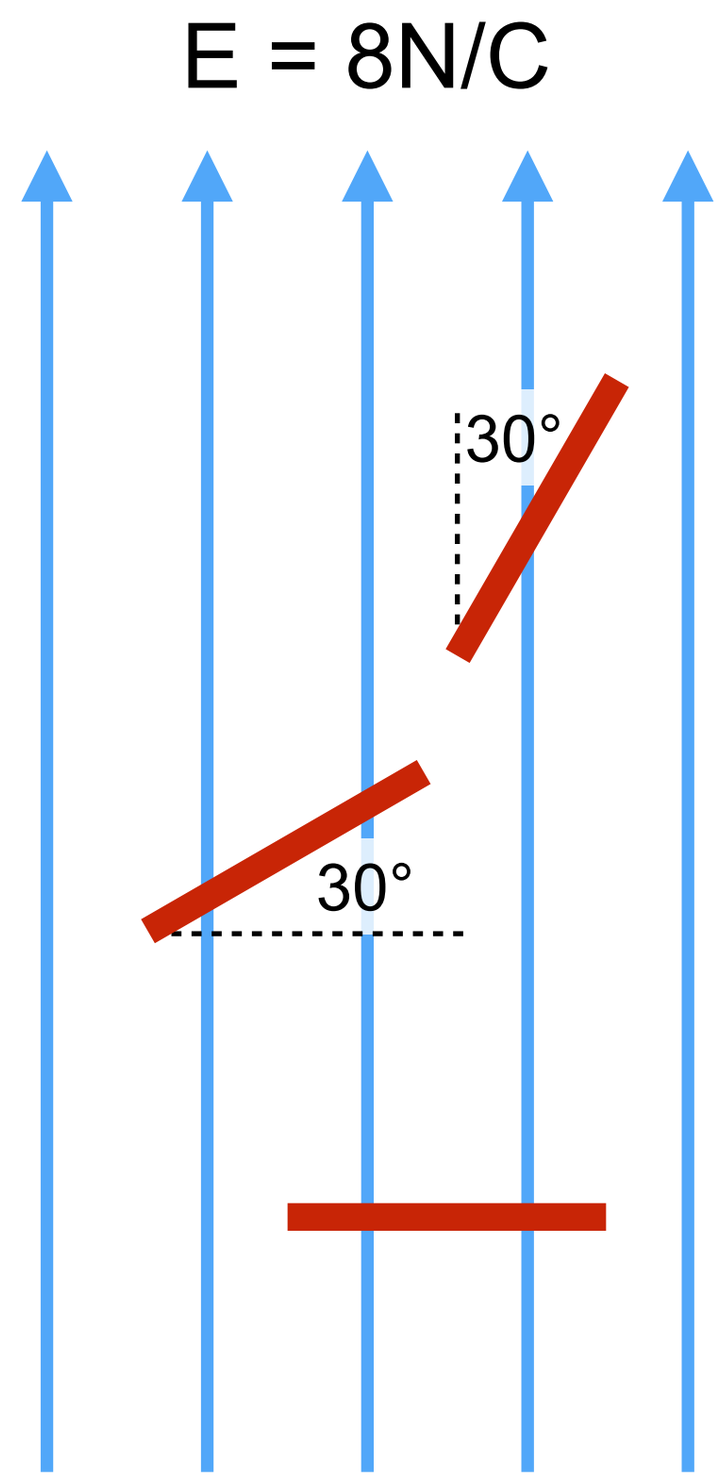
$= 1.4\ \mathrm{Nm^2/C}$



Bot : $\Phi_E = \vec{E} \cdot \vec{A}$

$= E\,A = E\,\pi\,R^2 = 1.6\ \mathrm{Nm^2/C}$





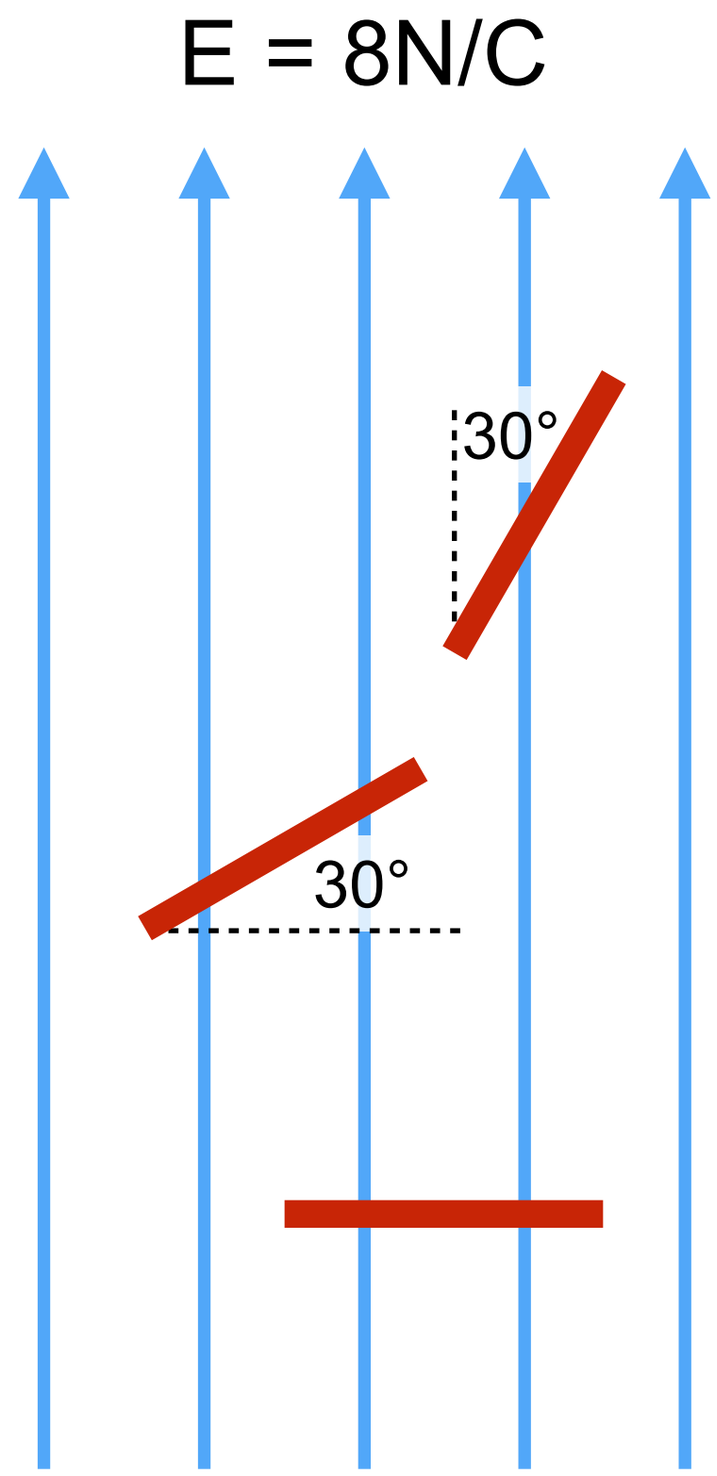
Example : Vertical Field

A uniform electric field points straight upwards with strength 8N/C.

Three circles with radius 0.25m are in the field, oriented as shown.

Which has the **smallest** flux through it?

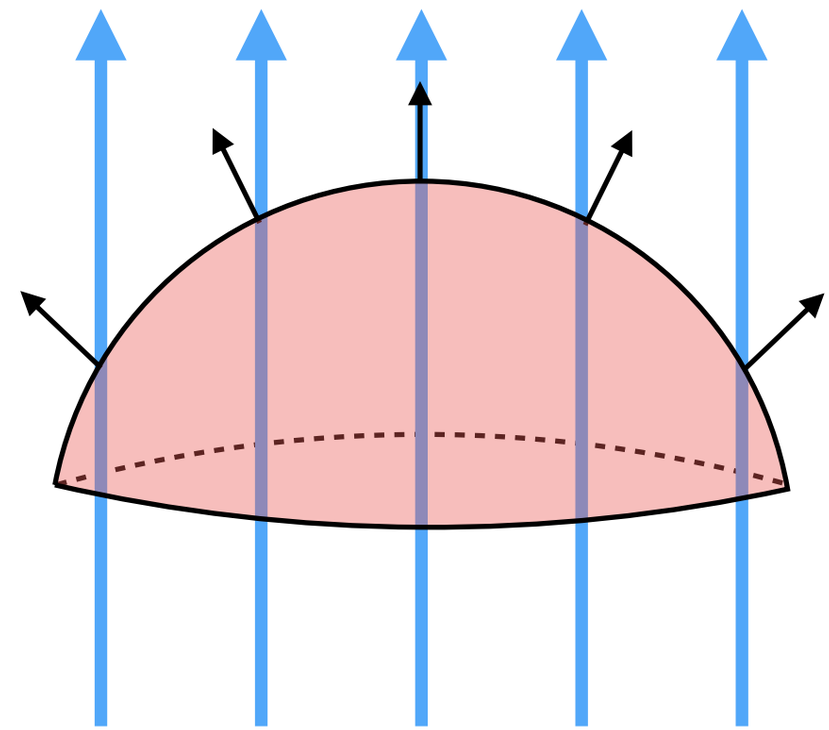
1. Top
2. Middle
3. Bottom
4. All the same
0. Don't wait on my response



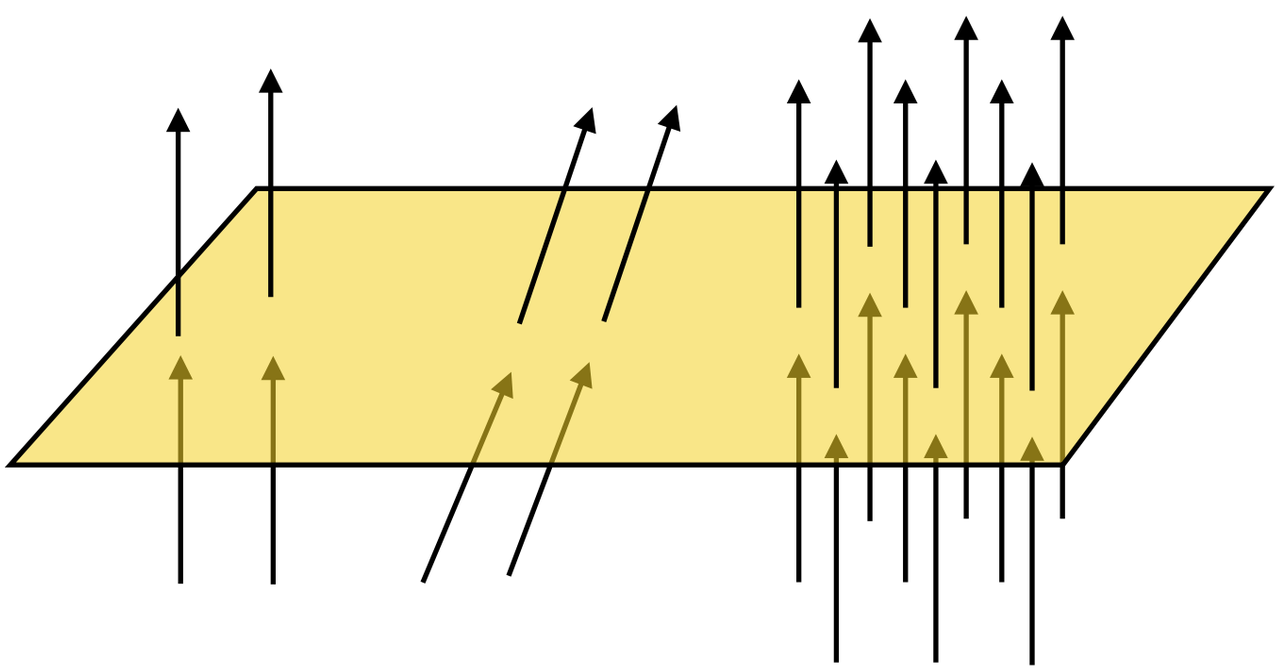
Surfaces & Flux

You can use <u>any</u> surface : even non-flat ones

$\vec{A}$ could point in different directions at different places on the surface.



What if the field is not uniform?!



|E| is different in different spots.

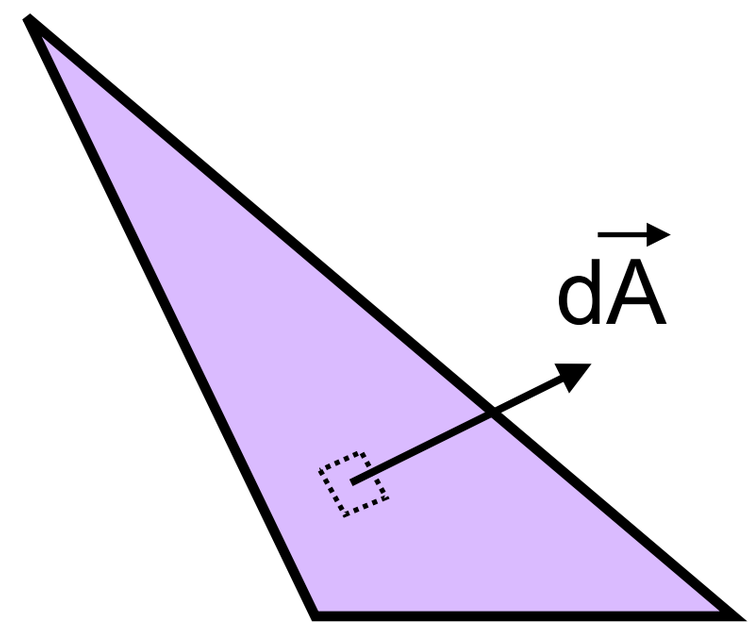
Direction of E changes across the surface.

Flux is an Integral

$$\Phi_E = \int \vec{E} \cdot d\vec{A} \longrightarrow \Phi_E = \vec{E} \cdot \vec{A}$$

If $\vec{E} \cdot d\vec{A}$ doesn't change across the surface

$d\vec{A}$ - differential area vector that is perpendicular to the surface, pointing out from it.

Magnitude : $\left| d\vec{A} \right| = dA$

$\int dA$ = Area of surface